Unsupervised Wavetable Seam Artifact Repair via Hybrid GA–PPO Meta-Search

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Julian M. Kleber
Independent Researcher
Email: julian.m.kleber@gmail.com

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Code (ReelSynth engine + DenoiseOpt meta-search): https://github.com/reeldemo/reelsynth, https://github.com/reeldemo/denoise-opt-meta

ABSTRACT

Looping a wavetable with mismatched endpoints produces a click at every period. We repair that wrap seam without paired clean recordings. **DenoiseOpt** searches small seam-repair operators (classical fades plus tiny residual networks) with a hybrid genetic-algorithm and reinforcement-learning loop. Candidates are ranked by a blended residual score that rewards seam fidelity to a no-cliff ideal sibling and mid-cycle identity to the cracked engine. Under the locked protocol, corrupt-to-corrupt Noise2Noise is the neural gate and Dual Cosine is the classical fade baseline. A 1,200-iteration hybrid run reaches about 0.970 on the holdout blend score but does not beat frozen Noise2Noise (about 0.975); Dual Cosine sits near 0.541. On twenty multi-family boards, Ours edges Noise2Noise on mean score and both crush Dual Cosine. On hard wrap cliffs, classical fades collapse while searched cells hold. The contribution is compact searchable bake cells that stay robust when classical fades fail and that need no paired clean targets at search time.

Keywords wavetable synthesis; wrap discontinuity; seam restoration; unsupervised learning; Noise2Noise; neural architecture search; reinforcement learning; genetic algorithms

1 Introduction

Wavetable oscillators store one period and wrap the read pointer at the end of the table. If the endpoints disagree ($x[0] \neq x[L-1]$), that wrap is a hard discontinuity. Under cyclic playback it becomes a click locked to the fundamental. The same geometry appears whenever a short loop is tiled for listening or scoring.

We treat the task as *cycle-local seam restoration*: edit a neighborhood of the wrap, and optionally a small residual cell over the period, so multi-period playback matches an ideal reference that never opens the wrap cliff. Agreement uses a discontinuity-weighted blend R_{blend} (seam fidelity vs the ideal, plus mid-cycle identity vs the cracked engine) under a latency-aware objective J (Section 3.3).

Four roles. Keep these distinct (Figure 1; Section 3.4):

1. **Ideal sibling** $r^{\star}$: cliff withheld. Scoring target only. Not a method.
2. **No-bake**: unrepaired cracked engine x, scored vs $r^{\star}$. A do-nothing reference.
3. **Dual Cosine**: classical raised-cosine end fade. Classical baseline.
4. **Ours (DenoiseOpt)**: searched bake cell from the hybrid outer loop.

Corrupt-to-corrupt **Noise2Noise** is a fifth comparator: the primary neural gate under the locked R_{blend} protocol (Section 5.2). It is not shown in Figure 1, which illustrates roles on one hard sine-wrap tile.

Classical virtual-analog tools already address wrap energy locally (BLIT/BLEP [1], PolyBLEP [3], BLAMP [2]). We score cycle-local bake versions of those operators beside Dual Cosine (Section 5.12). Label-free restoration such as Noise2Noise shows that useful reconstructors can be ranked without paired clean targets when the evaluation geometry is explicit [4, 5]. On the search side, NAS, aging evolution, PBT, CMA-ES, and evolution-guided policy gradients motivate a hybrid outer loop: GA for diversity, PPO for discrete architecture edits, and PBT-style exploit–mutate on elites [16, 19, 21–23].

Key findings. Under the locked v10.1 R_{blend} protocol:

- The hybrid champion does *not* clear the frozen Noise2Noise holdout gate (R_{blend}≈0.9697 vs 0.9750). Dual Cosine on that holdout sits near 0.541.

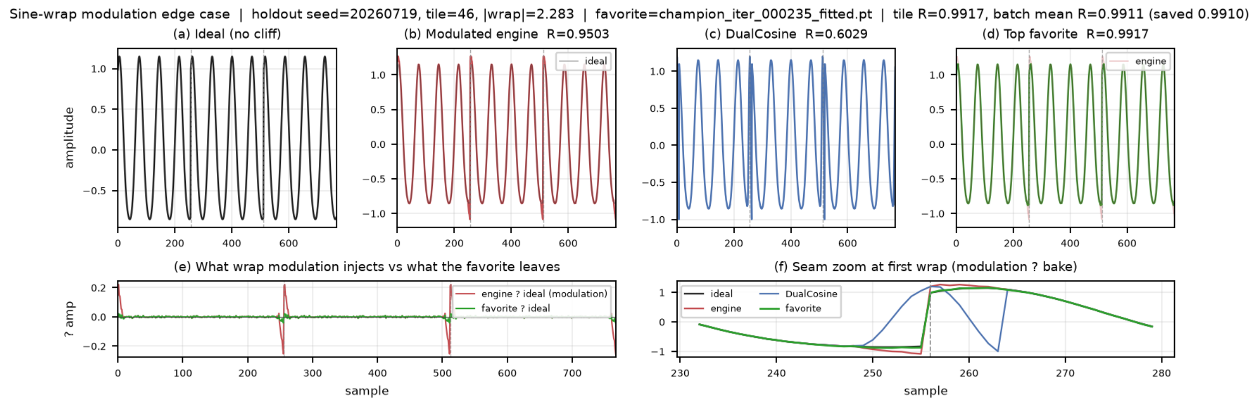

Figure 1: Hard sine-wrap holdout tile (seed `20260719`). (a) Ideal sibling $r^{\star}$. (b) No-bake cracked engine. (c) Dual Cosine fade. (d) Ours (searched bake). (e)–(f) Residual and seam zoom. Printed scores use whole-curve prolonged R for this figure; primary selection uses R_{blend} (Table 4).

- On twenty multi-family boards, Ours edges Noise2Noise on mean score (0.931 vs 0.925; 13/20 wins). Both crush Dual Cosine.
- On hard wrap cliffs, classical fades collapse while searched cells and Noise2Noise hold (Section 5.8).
- On Factory+FX and AKWF corpora under R_{blend}, Noise2Noise leads Ours; both still beat Dual Cosine.

Contribution in one line: compact searchable bake cells that remain robust when classical fades fail, and that need no paired clean targets at search time. Older whole-curve scores near $R{\approx}0.991$ are historical context only; they must not dominate the mental model of the locked protocol.

Transfer datasets. As a cross-domain stress test (not a diagnosis claim), we reuse the same wrap protocol on six boards (Section 4.2): **CWRU** and **MFPT** bearing vibration (one shaft revolution as a period); **MIT-BIH** and a **PTB-XL** 500 Hz ECG subset (one heartbeat as a period); plus **synthetic CNC** and **synthetic PMU/AC** pilots when real downloads are walled. Domain-trained Noise2Noise quality was not run there.

Scope. Cycle-local wavetable seam repair for DSP / synthesis venues. Caveats and exclusions live in Section 9.

1.1 Contributions

1. **Problem framing.** Wrap discontinuity as cycle-local seam restoration under tiled evaluation, with R_{seam}, R_{body}, and primary R_{blend}.
2. **Hybrid DenoiseOpt loop.** RL+GA(+PBT) search over bake graphs and continuous parameters, scored by R_{blend} and ranked by J.
3. **Open evaluation path.** Frozen Noise2Noise gate, multi-family and wavetable-native boards, and named transfer benches with released code and artifacts.

2 Related Work

We group prior work by mechanism. **Used** citations are anchors for residual-scored RL+GA meta-search. **Screened** citations were checked as contrast and are not dependencies.

Discontinuity control and modular DSP. BLIT/BLEP [1], PolyBLEP [3], and BLAMP [2] state the wrap and corner problem for trivial oscillators. We score cycle-local approximations beside Dual Cosine (Section 5.12). DDSP frames DSP modules as composable blocks [6]; we use that framing for small bake/FIR/MLP seam cells, not end-to-end vocoders.

Label-free restoration. Noise2Noise learns restorers from corrupted observations alone [4, 5]. Mixture-invariant training ranks unsupervised systems without stem labels [29]. We train a corrupt-to-corrupt seam CNN on the same generator as the primary neural baseline (Section 5). Full Demucs / DiffWave stacks are screened: domain and bake-budget mismatch [34, 37].

Compact waveform cells. Wave-U-Net, Conv-TasNet, dual-path, and attention audio models supply block priors we shrink to bake width [8, 11–15, 30]. They are design priors, not frozen full-scale separators inside every meta trial.

Search and hybrids. NAS taxonomies, RL controllers, progressive and latency-aware NAS, aging evolution, PBT, CMA-ES, and evolution-guided policy gradients motivate the hybrid outer loop [16–19, 21–24, 27, 28]. MoE soft gates motivate routing among small heterogeneous cells under one residual score [25]. Matched outer-loop bake-offs (Random, CMA-ES, REINFORCE, aging evolution, TPE vs hybrid) sit in Appendix B.

Used vs screened (short). Used: VA discontinuity [1–3]; unsupervised restoration [4, 5, 29]; DDSP [6]; compact waveform cells as above; NAS/RL/PPO/PBT/CMA-ES/ERL/MoE as above. Screened as contrast only: MAML [32], DARTS [33], Demucs-scale enhancers [34, 36], DiffWave [37], SEGAN-style full GAN loops [38]. Claims

Table 1: Core symbols used in Methods.

Symbol	Meaning
L	Cycle length (samples per period)
x	Cracked engine wavetable period (cliff on)
$y = \Theta(x;h)$	Baked cycle after seam ops / residual cell
Θ	Bake operator $\mathbb{R}^L \times \mathscr{H} \to \mathbb{R}^L$
$h \in \mathscr{H}$	Bake-cell configuration (ops + continuous θ)
$r^\star$	Ideal sibling (same seed; cliff withheld)
G	Procedural period generator (cliff on/off)
N	Prolong tiling count
W	Seam neighborhood width
R	Prolonged residual score in $[0,1]$ (1 = best)

stay inside cycle-local seam restoration under a declared compute budget.

3 Methods

3.1 Notation

Throughout, $x \in \mathbb{R}^L$ is the cracked single-period wavetable (engine input with open-wrap cliff). Let $\mathscr{H}$ be the space of bake-cell configurations (discrete operator graph plus continuous bake parameters). The bake operator is

$$\Theta : \mathbb{R}^L \times \mathscr{H} \to \mathbb{R}^L, \qquad y = \Theta(x;h), \quad h \in \mathscr{H}. \tag{1}$$

When continuous parameters alone are emphasized we write θ for the continuous slice of h and $y = \Theta(x;\theta)$. $r^\star \in \mathbb{R}^L$ is the procedural ideal sibling defined below. Tiling length N (typically N=16) forms $y_{\text{tiled}} = \text{Tile}(y,N)$ and $r^\star_{\text{tiled}} = \text{Tile}(r^\star,N)$. Seam width W (code: `SEAM_W`, typically 8) is the neighborhood edited by classical fade and polish ops. $R \in [0,1]$ is the prolonged residual score (1 = best). These symbols follow a different convention from Noise2Noise clean/noisy pairs.

3.2 Problem setup: cracked period and ideal sibling

Problem. Given a cracked period x, choose a bake configuration h so that the repaired cycle $y = \Theta(x;h)$ matches a no-cliff ideal under prolonged tiling.

Ideal sibling. Let $G(\text{seed}, \text{cliff})$ denote the procedural generator used by the CUDA FitCell batch maker (`make_batch` in the CUDA search runner): one draw of frequency, phase, and mid-cycle content, with an optional open-wrap cliff of width W at both ends. For any seed,

$$r^\star = G(\text{seed}, \text{cliff=off}), \qquad x = G(\text{seed}, \text{cliff=on}). \tag{2}$$

Engine and ideal therefore share the mid-cycle content of one draw and differ by the seam cliff (plus any seam-boosted noise applied only on the engine path). The ideal is *not* a latent studio ground truth and is *not* a method output. For an arbitrary cracked period from this family, the sibling is obtained by replaying the same seed with the cliff withheld. That is exactly the pair returned by `make_batch`. This sibling relationship makes unsupervised ranking possible without paired studio recordings [4, 5].

3.3 Blended residual score and latency-aware objective

Primary metric. After tiling length N, define unit residual scores on index masks. R_{seam} compares y to $r^\star$ on wrap neighborhoods of width W at each period head/tail. R_{body} compares y to the cracked engine x on the complementary mid-cycle body (identity prior: do not morph content away from the engine). The primary score is

$$R_{\text{blend}} = \alpha R_{\text{seam}}(r^\star, y) + (1-\alpha) R_{\text{body}}(x,y), \quad \alpha = 0.7. \tag{3}$$

with each component in the same clamp form $\text{clamp}(1 - \text{rms}/\text{rms}_{\text{ref}}, 0, 1)$. Whole-curve prolonged R remains a debug key only.

Search objective. Candidates maximize latency-aware

$$J = R_{\text{blend}} - \lambda \cdot \text{latency\_norm}, \quad \text{latency\_norm} = \frac{\log(1+t_{\text{ms}})}{\log(1+50)}, \quad \lambda = 0.02. \tag{4}$$

FitCell minimizes $1 - R_{\text{blend}}$; selection and champion updates use J. A champion must also strictly beat the frozen corrupt→corrupt Noise2Noise holdout R_{blend} (primary baseline).

Every scored row applies some Θ to x and is evaluated under (3). The ideal sibling is the seam scoring target; body identity uses the engine. We do not claim convergence guarantees for the hybrid outer loop, nor a general wrap-closure theorem for arbitrary bake cells.

Remark 1 (Noise2Noise primary baseline). `SeamN2N` (≈53.5k) is the primary neural comparator and champ gate. Search may discover `n2n_unet` cells at the same scale; TinyUNet1D `unet` is a different, smaller block [4, 5].

3.4 Seam bake operator and classical baselines

A **bake cell** $\Theta(\cdot;h)$ composes classical seam ops (DualCosine, polish, soft seam, FIR blends) with optional tiny residual cells (MLP/FIR/U-Net-lite/attention-lite/LSTM-lite) and optional MoE soft gates over experts (Appendix A, Algorithms 2–4). Bake parameters live in a bounded box; architecture strings are discrete graphs over those ops. Separate classical controls apply cycle-local BLIT/BLEP, PolyBLEP, and BLAMP residual corrections at the wrap on the cracked engine tile (same interface as DualCosine / FIR; not a full oscillator rewrite). Plain-language roles: BLIT/BLEP and PolyBLEP correct a *value* step at wrap; BLAMP corrects a *slope*/corner jump; DualCosine fades endpoints without a BLEP residual. They are scored in Section 5.12.

Four roles (keep separate).

- **Ideal sibling** $r^\star$: cliff withheld; scoring target only (panel (a) in Figure 1).
- **No-bake** (passthrough): $\Theta_{\text{no-bake}}(x) = x$; unrepaired cracked engine scored vs $r^\star$ (panel (b)). Not $r^\star$, not a residual-net skip, not a learned identity map.

- **DualCosine**: classical raised-cosine end fade on x (panel (c)); one classical board row and, separately, a PPO advantage baseline (Remark 2).
- **Ours**: searched bake $y = \Theta(x; h^\star)$ from the hybrid outer loop (panel (d)).

On mild cliffs residual mass $\ll$ ideal RMS under the unit clamp form of (3), so no-bake R_{blend} can sit high while the wrap click remains audible. That is a "do nothing" reference, not a perceptual wrap score. Released JSON may still use the legacy key `identity` for no-bake; manuscript prose uses *no-bake*.

Classical (non-AI) comparison board. Learned / searched methods are always ranked against the full classical board on absolute R (same $r^\star$): no-bake, DualCosine / cosine fade, `seam_fir3`, classic quadratic, hann/crossfade/soft fades, detrend/ensemble, poly seam, and cycle-local BLIT/BLEP / PolyBLEP / BLAMP where scored. When tables show ΔR vs DualCosine, that is a reporting gap vs one classical bake, not the definition of reward.

Remark 2 (DualCosine centering is PPO advantage only). The outer objective remains (4): maximize absolute $R(y, r^\star)$. In the PPO branch only, we optionally form a centered advantage

$$\rho = R - R_{\text{DualCosine}}, \tag{5}$$

so early-search advantages are roughly zero-mean. This does *not* change selection, FitCell loss, champion ranking, or the matched 5k meta-approach objective. All of those still maximize absolute R. DualCosine centering is soft relative shaping for the actor–critic, not a hard absolute ranking target.

3.5 Search pipeline

Figure 2 summarizes the residual-scored hybrid loop. Each outer iteration proposes a bake configuration h, runs FitCell (Appendix A, Algorithm 5) to fit continuous parameters under loss $1 - R$, evaluates prolonged R vs $r^\star$, and updates the population / PPO policy. Pseudocode for GA, PPO, PBT, and the full hybrid rotation is in Algorithms 6–9.

3.6 Search space and inner fit

Each genome encodes: ordered op list (fade, polish, FIR, pin, residual write), depth, residual cell type (none, MLP, FIR, U-Net-lite, attn-lite, LSTM-lite), optional MoE expert count $K \in \{2, 3, 4\}$, and continuous θ in the CUDA runner box bounds. The searchable bake vocabulary includes a width-capped 1–2 layer LSTM block over the seam window or full cycle (distinct from the fixed supervised seq-LSTM SOTA baseline). The live hybrid tag is `PPO+GA+PBT+NAS+depth+MoE` with LSTM enabled in the block graph.

FitCell draws sibling batches via G, applies $\Theta(\cdot; h)$, and steps Adam on $1 - R$ with early stopping on plateau (Algorithm 5).

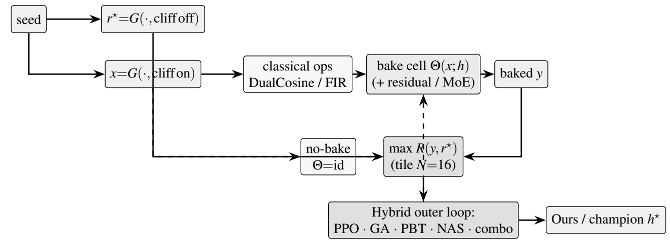

Figure 2: DenoiseOpt overview (before component deep-dives). Same-seed generator G yields ideal sibling $r^\star$ (cliff off) and cracked engine x (cliff on). Bake cell $\Theta(x; h)$ may include DualCosine/FIR and optional residual/MoE experts. No-bake is the identity control $x \mapsto x$. Outer objective is maximize absolute prolonged $R(y, r^\star)$. DualCosine centering (not shown) is PPO advantage shaping only. Dashed feedback: architecture proposals update the cell. Pseudocode in Appendix A.

3.7 Hybrid outer loop

Each iteration rotates among branches: PPO, GA, PBT, NAS, and combo. Selection and FitCell are always by absolute residual R (maximize toward the ideal sibling; loss $1 - R$).

Near-ceiling scores need reward shaping. On the sine+cliff generator, no-bake already sits near $R \approx 0.97$, and strong classical / searched bakes live in a narrow band toward 1 (often 0.99–0.991). Absolute gaps of order 10^{-3}–10^{-2} are real but tiny as raw PPO/GA signals. Without shaping, advantages collapse. Hyperparameters (clip, entropy, mutation rate, population size, fit steps, learning rate) then decide whether the outer loop can climb the last percent. We therefore treat reward shaping and search hyperparameters as first-class tunables alongside bake architecture. In the present hybrid, PPO uses a searchable shaping mode among {`abs_r`, `vs_dualcosine`, `vs_nobake`, `neglog_gap`} (PBT-mutated; default $\rho = R - R_{\text{DualCosine}}$ per Remark 2). PBT mutates fit / PPO hyperparameters and reward mode in-population; plateau adaptation retunes branch mix and depth/MoE bias when champions stall. The matched 5k meta-approach comparison (Section 3.8) is the sole GPU experimentation vehicle for outer-loop family, HP, and reward-shaping sweeps under a shared FitCell budget.

3.8 Meta-learning approach comparison

Beyond the hybrid outer loop, we compare five additional outer-loop families under a matched 5,000-evaluation budget, shared search seed `1902771841`, and the same FitCell / prolonged-R protocol (LSTM and xLSTM included in the searchable bake vocabulary). Table and figure legends use the short display names below (artifact folders keep stable code keys; see the nomenclature):

- **Random NAS** (`random`): independent uniform draws over bake graphs and fit hyperparameters (control).
- **Cont. CMA-ES** (`cmaes`): diagonal covariance-matrix adaptation over a continuous encoding of depth, width, cell kind, block logits, and fit hyperparameters [24].

- **Arch REINFORCE** (`reinforce`): vanilla policy gradient over discrete architecture-edit actions (no clipped surrogate; contrast to PPO [17, 19]).
- **Aging evolution** (`aging_evo`): regularized evolution with oldest-member replacement after tournament parent selection [20, 21], distinct from the hybrid's non-aging tournament GA branch.
- **TPE Bayes NAS** (`tpe`): lightweight tree-structured Parzen estimator over discrete bake categoricals in the spirit of practical Bayesian optimization [26].

We also run **Ours (hybrid GA–PPO)** (`hybrid_lstm`): the multi-branch hybrid loop for 5,000 evaluations with LSTM/xLSTM in the block vocabulary and online reward-mode / fit–PPO hyperparameter co-tuning, as the proposed matched-budget arm. Results appear in Appendix B (Table 15, Figures 19–20); healed-sample overlays in Figure 6.

Remark 3 (Trial complexity). Per outer iteration the dominant cost is $O(T_{\text{fit}} \cdot C_{\text{fwd}})$ for fit steps times one forward bake. Bake-width caps (tiny residual cells) keep C_{fwd} far below full-band Demucs/DiffWave forwards screened in Related Work.

3.9 Hyperparameters

Table 2 lists the CUDA search defaults used for the 5k-gate freeze reported in Results. Under the near-ceiling residual regime (Section 3.7), these outer-loop and fit hyperparameters are co-tuned with architecture; PBT and plateau adaptation continue to mutate several of them online. Companion listings: Algorithms 1–9 (Appendix A) and the pseudocode notes.

Table 2: Hybrid search hyperparameters (RTX 3090 DenoiseOpt CUDA search-runner defaults). Co-tuned with reward shaping under near-ceiling R (Section 3.7).

Parameter	Value
Population size n	12
GA tournament k	3
GA crossover fraction p_c	0.5 (else clone elite parent)
GA mutation	≥ 1 mutate after inherit (+50% second mutate)
PPO clip ε	0.2 (default; searchable $\{0.1, 0.2, 0.25, 0.3\}$)
Adam learning rate (fit)	3×10^{-3} (default; PBT-searchable)
Adam learning rate (policy)	3×10^{-4}
Entropy coefficient (PPO)	0.02 (default; searchable $\approx$ 0.005–0.06)
PBT exploit / mutate	elite fraction 0.25; multi-step arch+hp mutate
MoE routing	`sequential` or `moe_parallel` when enabled
Plateau boredom threshold	1000 iters without champ improve
Cycle length L / tiles N / seam W	256 / 16 / 8
Fit steps / batch (default)	24 / 48
Algo tag	`PPO+GA+PBT+NAS+depth+MoE`

4 Experiments

Locked evaluation protocol. Evaluation follows the v10.1 evaluation protocol. Primary metric: R_{blend} (α=0.7). Search objective: J as in (4) (λ=0.02). Champ gate: strictly beat frozen Noise2Noise corrupt→corrupt holdout R_{blend}. Secondary: SNR, SDR, wrap-jump, seam-local edge RMSE; debug keys include pure R_{seam} and whole-curve R. Holdout seed `20260719`. Search seed `1902771841`. N2N train seed `424242`. We report the hybrid_lstm freeze plus holdout / multi-family / Factory+FX / AKWF / transfer benches. PESQ/STOI stay out of scope on non-speech cycles. LibriSpeech and MUSDB are not used. Venue: DSP / synthesis artifact repair (DAFx / AES / arXiv cs.SD).

4.1 Dataset

Creation. The paper-facing evaluation corpus is a frozen draw from the same synthetic generator used by the CUDA search runner (`make_batch`). Each cycle has length L=256. A two-harmonic sine is drawn with frequency $\sim \mathscr{U}(1,4)$ and phase $\sim \mathscr{U}(0,2\pi)$. An open-wrap seam cliff of amplitude $\pm\mathscr{U}(0.08, 0.43)$ is applied over `SEAM_W`=8 samples at both ends. Seam-boosted Gaussian noise (scale 0.02 at the ends, $0.15\times$ in the mid-cycle) is added to form the engine input. The ideal withholds the cliff and is used only as the scoring target $r^\star$ (never as a method output). No-bake leaves the cracked engine unchanged and scores that unrepaired x against the same $r^\star$. Prolonged residual R tiles each cycle N=16 times before the RMS ratio. The holdout seed is `20260719` (distinct from the hybrid search seed `1902771841`). The search campaign draws fresh i.i.d. batches from this generator and does not train on the frozen holdout tensors. There is no supervised clean/noisy training split of studio audio.

Multi-family diversity ($\geq$ 20 waveforms). Beyond the single sine+cliff holdout, Phase 3a scores methods on **20** generative draws: ten family variants (`sine_cliff`, `harmonic_fft`, `am_fm`, `nonlinear`, `combo`, `triple_mix`, `extreme_overlay`, `open_wrap_bias`, `soft_noise`, `wide_seam`) $\times$ two seeds (`20260719` and `20260720`), batch 64 each (128 cycles per family; 1,280 total). These Python generative variants stress harmonic / AM–FM / nonlinear / overlay / open-wrap behavior of `make_batch` and remain the primary SOTA matrix. Separately, `export_sound_bench_tiles` exports **20** Rust `sound_bench` engine/ideal pairs (10 families $\times$ 2 seeds, L=256, byte-aligned with `generate_sound` / `generate_sound_ideal`) scored under the same residual geometry as a secondary transfer block (Table 12). Every method sees the same tensors per waveform. **Primary classical comparison** is absolute prolonged R against the non-AI board (no-bake, DualCosine, `seam_fir3`, poly, fades, VA residuals; Section 3.4). The search objective is

Table 3: Dataset statistics (evaluation boards). L=256 unless noted. Hours are raw source duration on disk where measurable, not tiled period count. Plot 4 compares ≈1,280-height boards only (multi-family / Factory+FX / AKWF / transfer).

Board	Size	Notes
Holdout metrics	4,096 cycles	seed 20260719; N=16 tiles
Score batch	64 cycles	classical / favorite tables
Multi-family SOTA	1,280 cycles	10 families × 2 seeds × 64
Factory+FX export	1,280 periods	640 dry + 640 FX mid-tile extracts
AKWF OA	1,280 periods	CC0 single-cycle WAVs
Meta hybrid_lstm	1,200 iters	$R_{\text{blend}}+J$; seed 1902771841
CWRU	256 periods	4 MAT files; DE @12 kHz
MFPT	256 periods	shaft-rate windows
MIT-BIH	256 periods	10 records @360 Hz
PTB-XL subset	256 periods	94 *_hr @500 Hz
synth CNC / PMU	256+256	procedural pilots

maximize R toward the ideal sibling (best R=1). DualCosine is one classical comparator and a PPO centering constant. It is not the reward target.

Dataset statistics. Table 3 and Figures 3–4 inventory every evaluation board used in this paper. The primary procedural holdout is a frozen 4,096-cycle metrics draw (L=256, N=16 tiles, seed 20260719) plus a matched score batch of 64 cycles (five consecutive seeds for classical mean±std). Multi-family SOTA scores 20 waveforms (10 families × 2 seeds, batch 64; 1,280 cycles). Wavetable-native boards use ReelSynth Factory+FX exports (n=1,280: 640 dry + 640 FX) and AKWF CC0 cycles (n=1,280) so Plot 4 corpus bars are comparable to multi-family (1,280). Sci/eng transfer tiles 1,536 periods across CWRU, MFPT (n=256), MIT-BIH, PTB-XL subset, and synthetic CNC/PMU. Search never trains on the frozen holdout; N2N/seq baselines use disjoint train seeds 424242/424243. Artifacts: dataset artifacts.

Dataset metrics. A metrics draw of 4,096 cycles under the holdout seed yields mean ideal RMS 0.721 ± 0.016, mean engine residual RMS 0.021 ± 0.013, and mean absolute wrap jump $|x_0 - x_{L-1}|$ of 0.913 ± 0.634. No-bake (passthrough) on that draw scores mean $R \approx 0.971$. Method tables use a matched score batch of 64 cycles from the same seed (the canonical evaluation tensors), five consecutive seeds for mean±std on the classical table, and the 20-waveform matrix for SOTA secondary metrics. Figure 5 summarizes the 4,096-cycle metrics draw: ideal RMS clusters near 0.72, wrap jumps span nearly three orders of magnitude (median ≈ 0.83), and no-bake R is high on easy tiles but falls below 0.91 on the hardest decile of wrap jumps. Figure 1 is the documented sine-wrap edge case: a high-$|wrap|$ holdout tile where cliff modulation is visible, scored with DualCosine and the top neural favorite under the same residual R (tile $R \approx 0.9917$, favorite batch mean $R \approx 0.991$).

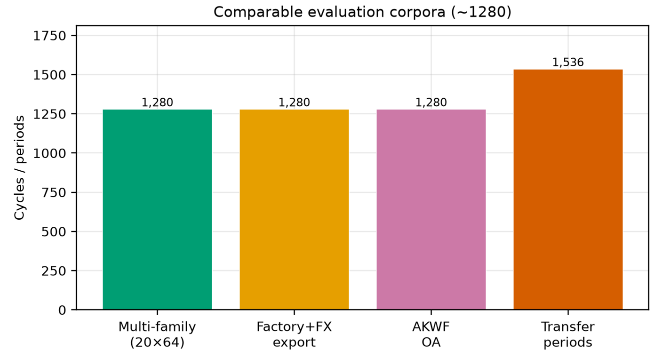

Figure 3: Corpus sizes across evaluation boards (cycles or periods). Holdout metrics draw, matched score batch, multi-family SOTA, factory/AKWF wavetable-native sets, and summed transfer periods.

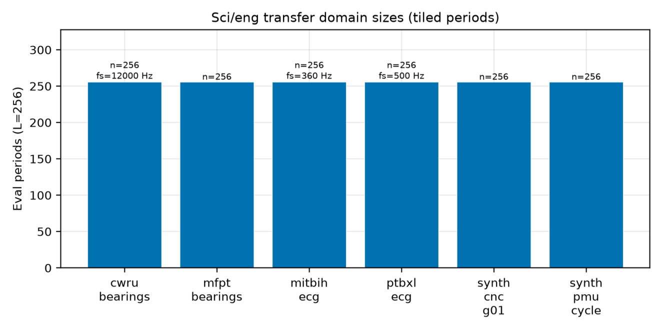

Figure 4: Sci/eng transfer domain sizes: tiled eval periods at L=256 (with source f_s where applicable).

Hybrid search campaign (5,000-iteration freeze). A CUDA search tagged PPO+GA+PBT+NAS+depth+MoE runs with a large declared iteration budget (order 10^5). At the reported freeze (run gpu-rl-arch-20260719T083019Z, ≈ 6,922 clean iters / ≈ 7.9 h on RTX 3090) the champion residual is $R \approx 0.99093$ against DualCosine baseline ≈ 0.81658 ($\Delta R \approx +0.174$). Branch-best residuals at the freeze: GA ≈ 0.99016, PBT ≈ 0.99012, PPO ≈ 0.98924, combo ≈ 0.98854, NAS ≈ 0.98677. Isolated short-budget re-runs (GA-only, PPO-only, GA+PPO, full, 150 iters, same search seed, plateau adapt off) are reported beside those freezes in Table 16. Figures in Section 5 regenerate from the archived search results.

Inference, classical, and learned baselines. High-R fitted cells ($R \geq 0.98$) are timed on CUDA (batch 64, prolonged residual). Ten non-learnable bake denoisers share the frozen canonical holdout against the neural favorite (Table 6). Cycle-local BLIT/BLEP, PolyBLEP, and BLAMP seam repairs are scored on the same holdout / cliff / multi-family protocol (Table 11; VA results) and illustrated on the

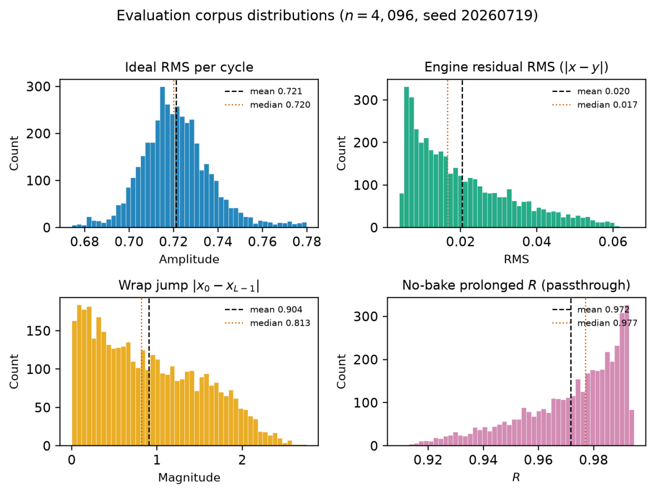

Figure 5: Holdout distribution statistics (seed `20260719`, n=4,096 cycles). Distinguish: ideal sibling $r^\star$ (cliff withheld); no-bake = unrepaired cracked engine (not $r^\star$); DualCosine = one classical bake. Top left: ideal RMS per cycle. Top right: engine residual RMS before any bake. Bottom left: wrap jump magnitude $|x_0 - x_{L-1}|$ (cliff hardness proxy). Bottom right: no-bake prolonged R (passthrough, $x \mapsto x$). Dashed and dotted vertical lines mark mean and median. Hard-cliff reporting (Section 5) uses the top 10% wrap-jump stratum where no-bake R drops.

intro high-wrap tile (Figure 10). Fixed-arch MLP-on-R and CNN/U-Net-lite-on-R baselines (no outer NAS) are trained on $1-R$ with the same generator and scored in the SOTA matrix. Phase B adds a primary corrupt→corrupt Noise2Noise seam CNN (train seed `424242`, disjoint from holdout), a secondary sibling-supervised ceiling on the same architecture, and lightweight LSTM / depthwise 1D-CNN seq restorers (train seed `424243`). No search-champion weights initialize those baselines. Family-conditional hardness uses the separate Rust procedural 100,000-cycle bench reported in Results.

Cliff strata and wavetable-native realism. A 4,096-tile holdout draw (seed `20260719`) is stratified by engine wrap-jump percentiles (top 25% / top 10%; cutoffs in the cliff-strata data). Hard-cliff is the operative regime for claims about DualCosine collapse vs favorite retention. Edge RMSE is the locked required seam-local secondary; click energy is optional. Wavetable-native transfer (Phase F1) scores true ReelSynth-exported factory periods (primary) and external AKWF CC0 WAVs (secondary) under the same close-seam→open-wrap protocol. Procedural stand-ins are tertiary smoke only. Speech/music LibriSpeech/MUSDB corpora remain off.

Audible demos and vibrato spectrograms. Downloadable clips (no-bake / DualCosine / Ours; 44.1 kHz mono, A4, 3 s): hear-sample WAVs. Rebuild script: `export_meta_hear_samples.py`. Slow-vibrato spectrogram protocol (`bench_vibrato_spectrogram.py`; artifacts: vibrato folder) scores the same bake cells under modulated-pitch playback; figures in Section 5.15. These assets support informal audition only. Formal MOS/MUSHRA scores were not collected; the planned listening protocol is in Section 5.19.

4.2 Cross-domain wrap transfer protocol

The primary claim of this paper is wavetable seam repair. As a stress test, we ask whether the *same* cycle-local wrap geometry and DenoiseOpt search still help on short periodic signals from other fields (not as domain diagnosis SOTA, but as "does a wrap cliff + ideal sibling still make sense?"

Shared transfer recipe. For each domain we cut fixed-length periods (L=256), build an ideal sibling (smooth / template / cubic path) and a cracked engine (same content with an open-wrap cliff, sometimes with a deliberately worse interpolant), then run the same prolonged residual R FitCell objective and `hybrid_lstm` outer loop used in the historical wavetable search freezes (150–250 outer iterations per dataset, `fit_steps` = 40). (This transfer pilot predates the v10.1 $R_{\text{blend}}+J$ lock; do not read Table 13 as R_{blend}.) Classical baselines (no-bake, DualCosine, FIR, endpoint pin, . . .) share that board. Domain-trained Noise2Noise quality was not executed here. Artifacts: signal_heal_transfer.

What each dataset is.

CWRU bearings. Public vibration recordings from the Case Western Reserve University Bearing Data Center (drive-end accelerometer, typically 12 kHz). One shaft revolution is treated as one "period": the ideal uses cubic resampling of the revolution; the engine uses a coarser linear path plus a wrap cliff (bad change-of-time proxy). Real data; n=256 periods.

MFPT bearings. Fault-bearing vibration from the Society for Machinery Failure Prevention Technology Fault Data Sets (fixed 25 Hz shaft rate in our windows). Same cubic-ideal / linear+cliff-engine construction as CWRU. Real data; n=256 periods.

MIT-BIH ECG. Classic PhysioNet arrhythmia ECG corpus. We take normal R–R beat windows as periods: the ideal is a local mean beat template with mild endpoint equalization; the engine is the beat plus an artificial wrap cliff. Real data; n=256 periods.

PTB-XL ECG (subset). Large PhysioNet clinical ECG collection. We use the 500 Hz `records500` / `*_hr` lead-I beat windows only (a subset pilot) (n=256 periods from 94 records), not the full PTB-XL corpus. Same beat-template ideal / cliffed engine idea as MIT-BIH.

Synthetic CNC G01. Stand-in for closed machine-tool contours (sharp corners vs a wide blend). Real KIT CNC downloads are login-walled, so this row is a procedural pilot with the same wrap protocol; n=256. Not a claim on industrial toolpaths.

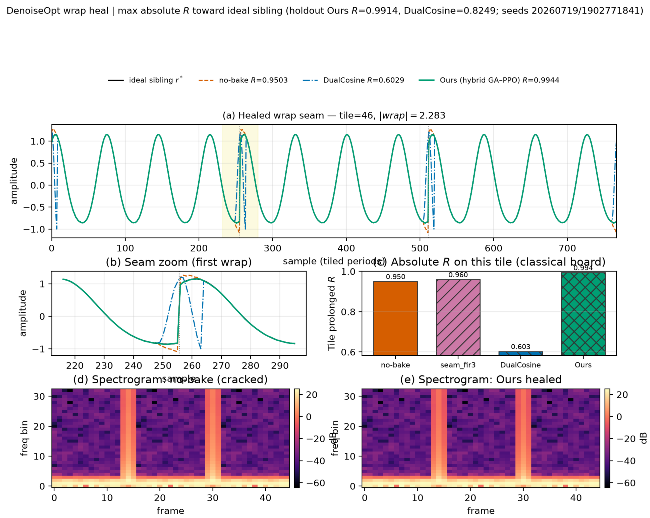

Figure 6: Holdout wrap-seam heal for the hybrid champion vs no-bake, Dual Cosine, and ideal sibling $r^\star$ (seed `20260719`).

Synthetic PMU / AC cycle. Stand-in for one power-system AC / PMU voltage cycle with a wrap cliff. Real IEEE DataPort PMU streams need an account, so this row is synthetic; n=256. Not a claim on live grid telemetry.

Scope note. Transfer tables are classical-board comparisons under this residual protocol. Deep diagnosis SOTA and formal listening are out of scope (Section 9). Results appear in Section 5.18.

5 Results

Reading guide. Primary numbers use locked R_{blend} (Sections 5.1–5.2). Hard-cliff classical collapse and multi-family ranking follow. Matched outer-loop bake-offs, HP probes, branch freezes, and transfer latency sit in Appendix B. Honesty notes are collected in Section 9.

5.1 Hybrid search under the blended score

Does hybrid GA+PPO beat the frozen Noise2Noise gate under R_{blend} and J? A 1,200-iteration search (seed `1902771841`, $\approx$1.65 h wall) selects on J (4) (λ=0.02). Champion: $R_{\text{blend}}{\approx}0.9695$, $J{\approx}0.9603$, $t{\approx}5.08$ ms. Search-monitor Dual Cosine on the same runner: $R_{\text{blend}}{\approx}0.504$. Frozen Noise2Noise holdout gate: $R_{\text{blend}}{\approx}0.9750$. The search champ does *not* clear that gate, while clearly clearing Dual Cosine. Artifacts: hybrid search data.

5.2 Noise2Noise vs Ours on four boards

Table 4 summarizes the four evaluation boards under R_{blend}, with Dual Cosine as the classical fade row on every board. On the frozen sine+cliff holdout (seed `20260719`), Noise2Noise leads Ours (0.9750 vs 0.9697; $\Delta{\approx}{-0.0053}$), while Dual Cosine collapses to 0.5409 (Ours−DualCosine $\Delta{\approx}{+0.429}$). On the 20-waveform multi-family board, Ours mean 0.9311 exceeds N2N 0.9249 (13/20 wins; Δ mean $\approx$+0.0062) and both dominate Dual Cosine (mean $\approx$0.514). On balanced wavetable-native corpora (n=1,280 each), N2N leads Factory+FX (0.771 vs Ours 0.699; Dual Cosine 0.589) and AKWF (0.731 vs Ours 0.679; Dual Cosine 0.424). Latency: holdout Ours $\approx$4.57 ms vs N2N $\approx$0.65 ms (J still favors N2N on holdout).

Table 4: Ours vs Noise2Noise and Dual Cosine under R_{blend} (α=0.7). Δ columns are Ours minus that baseline. Gate: Ours exceeds N2N.

Board	Ours	N2N	DualCosine	Δ_{N2N}	Δ_{DC}	Gate
Holdout	0.9697	0.9750	0.5409	−0.0053	+0.4289	no
Multi-family (mean)	0.9311	0.9249	0.5144	+0.0062	+0.4167	13/20
Factory+FX	0.6990	0.7714	0.5887	−0.0724	+0.1104	no
AKWF OA	0.6794	0.7305	0.4244	−0.0511	+0.2551	no

Earlier whole-curve scores (context only). Earlier whole-curve prolonged-R freezes ($R{\approx}0.991$ vs DualCosine $\approx$0.825) are historical context only; v10.1 selection uses $R_{\text{blend}}{+}J$ and Noise2Noise as the primary neural baseline.

5.3 Which families stay hard

On the fitted DenoiseOpt 100,000-cycle bench, denoise score D is lowest for `nonlinear` ($D \approx 0.39$) and `combo` ($D \approx 0.40$), then `triple_mix` / `extreme_overlay` ($D \approx 0.51$–0.52), while `harmonic_fft` and `am_fm` sit near $D \approx 0.69$–0.70. Lit-combo champion family stress (prolonged residual) similarly ranks `open_wrap_bias` ($R \approx 0.83$) and `extreme_overlay` / `combo` ($R \approx 0.88$–0.89) below `harmonic_fft` ($R \approx 0.95$). Hard sounds recur.

Earlier whole-curve search freeze. At the reported 5k-gate search freeze ($\approx$ 6,922 clean iterations, $\approx$ 7.9 h on RTX 3090) the champion residual was $R \approx 0.99093$ with DualCosine ≈ 0.81658 on the CUDA search runner ($\Delta R \approx +0.174$). That freeze used whole-curve prolonged R (superseded by R_{blend} for v10.1 selection). Monitoring plots below are regenerated from that freeze.

5.4 Inference latency at matched score

Across diverse high-R fitted cells ($R \geq 0.98$), we timed CUDA forward passes (batch 64, prolonged residual). Within a tight band of the best residual ($R \geq R_{\max} - 5 \cdot 10^{-4}$), the **favorite** is the lowest-latency model: `champion_iter_000235` with $R \approx 0.9910$, $\approx$ 3.15 ms/batch, $\approx$ 37k parameters (versus a higher-capacity max-R cell at $R \approx 0.9914$ but $\approx$ 7.18 ms/batch / $\approx$ 104k params). Selection rule: maximize R, then minimize latency inside the band.

5.5 Top-5 architectures

Table 5 ranks five distinct fitted bake graphs from the CUDA inference bench (inference data, batch 64), preferring residual R first and latency on ties, and skipping near-duplicate copies of the same topology signature.

5.6 Classical methods vs neural favorite

Table 6 scores the classical (non-AI) bake set (no-bake, DualCosine, and `seam_fir3`) plus the neural favorite on

Table 5: Top-5 distinct fitted architectures on the CUDA inference bench (batch 64, prolonged R, seed geometry matching the search campaign). Favorite is latency-best inside the near-max-R band ($R \geq R_{\max} - 5 \cdot 10^{-4}$).

Tag	R_{saved}	R_{live}	ms/batch	params
`iter_000157`	0.9914	0.9913	7.18	104 484
`iter_000205`	0.9911	0.9907	7.29	70 562
`iter_000235*`	0.9910	0.9915	3.15	37 489
`iter_000397`	0.9910	0.9907	6.80	97 262
`iter_000229`	0.9909	0.9902	6.67	100 500

*Favorite (fastest in $R \geq R_{\max} - 5 \cdot 10^{-4}$ band).

Table 6: Canonical holdout under historical prolonged R (seed `20260719`). Mean±std over five seeds vs the same $r^\star$. No-bake = unrepaired engine.

Method	R	R (5-seed)	ms/batch
neural favorite	0.9911	0.9912 ± 0.0002	2.56
no-bake (passthrough)	0.9718	0.9726 ± 0.0021	0.31
`seam_fir3`	0.9610	0.9616 ± 0.0019	0.67
classic quadratic	0.8449	0.8453 ± 0.0117	1.59
DualCosine	0.8249	0.8258 ± 0.0137	1.35
cosine fade	0.8249	0.8258 ± 0.0137	1.59
crossfade	0.8116	0.8130 ± 0.0145	1.50
hann blend	0.8048	0.8060 ± 0.0156	0.54
soft wide fade	0.7432	0.7378 ± 0.0171	2.64
detrend	0.4079	0.4091 ± 0.0386	0.36
ensemble detrend+DC+FIR	0.4024	0.4027 ± 0.0387	1.56

the frozen corpus of Section 4.1 (CUDA, batch 64, seed `20260719`). Every row uses the same ideal sibling $r^\star$; higher R means closer to that sibling. DualCosine reaches $R \approx 0.825$ (≈ 1.35 ms/batch), near the search-monitor DualCosine ≈ 0.817 on live draws from the same generator. The best *active* classical repair is seam-local 3-tap FIR (`seam_fir3`) at $R \approx 0.961$ / ≈ 0.67 ms/batch (0.962 ± 0.002 over five seeds). No-bake (passthrough; unrepaired engine vs $r^\star$) scores $R \approx 0.972$. It is a do-nothing reference, not the ideal. The neural favorite reaches $R \approx 0.991$ (0.991 ± 0.0002 over five seeds) at ≈ 2.56 ms/batch (≈ 37k params): $\Delta R \approx +0.019$ vs no-bake, $\Delta R \approx +0.030$ vs `seam_fir3`, and $\Delta R \approx +0.166$ vs DualCosine. This table is a classical-board snapshot under prolonged R; Noise2Noise appears on the multi-family matrix (Table 7: favorite 0.977 vs N2N corrupt→corrupt 0.970 vs DualCosine 0.814) and on the R_{blend} four-board (Table 4).

5.7 Multi-family comparison

Does the meta favorite beat the classical non-AI board (no-bake, DualCosine, FIR, poly, fades), fixed MLP/CNN baselines, *and* Noise2Noise across twenty generative families? Table 7 reports mean±std over the 20-waveform multi-family set (Section 4.1), with canonical-holdout SNR/SDR/wrap-jump for the same method set from the SOTA matrix. Figures 7–8 plot the same ranking. Primary reading is absolute

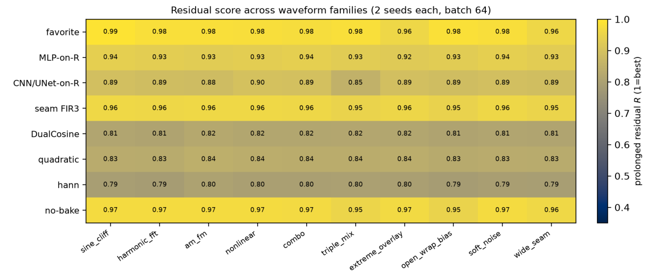

Figure 7: Colorblind-safe (`cividis`) heatmap of mean prolonged R for selected methods × generative families (2 seeds each). Higher is better.

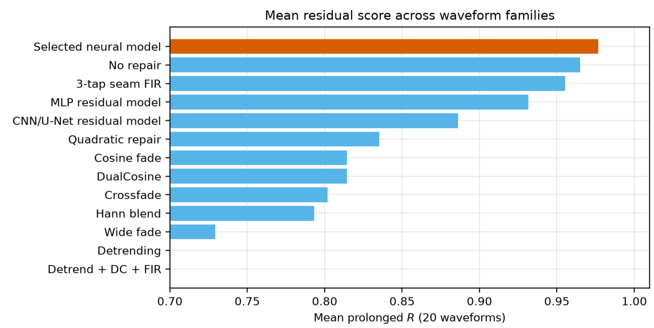

Figure 8: Multi-family mean prolonged R by method (same 20-waveform board as Table 7).

prolonged-R rank (historical board; see Limitations): neural favorite 0.977, corrupt→corrupt N2N 0.970, no-bake 0.965, poly 0.966, `seam_fir3` 0.955, DualCosine 0.814 (historical whole-curve board; not the v10.1 R_{blend} four-board in Table 4). Fixed MLP-on-R and CNN/U-Net-lite-on-R (no outer NAS) sit between DualCosine and the meta favorite. Paired Wilcoxon signed-rank (normal approx.) of favorite vs DualCosine on the 20 waveforms: $W{=}0$, $p{\approx}8.9{\times}10^{-5}$, mean $\Delta R{=}{+}0.162$ (bootstrap 95% CI $[0.155, 0.170]$); vs no-bake the multifamily mean gap is smaller ($\approx +0.012$) but still favors the searched bake on this generator. Vs corrupt→corrupt N2N the mean gap is smaller still ($\approx +0.007$ on this prolonged-R board).

5.8 Hard cliffs expose classical fade failure

Does high no-bake R hide DualCosine failure on large wrap jumps? Table 8 stratifies a 4,096-tile holdout draw (seed `20260719`) by engine wrap-jump percentiles (p75≈ 1.387, p90≈ 1.832; frozen in the cliff-strata data). This board uses whole-curve prolonged R (historical; not R_{blend}). No-bake R can stay high on mild cliffs because residual mass remains small relative to ideal RMS. DualCosine collapses on hard cliffs ($0.819 \rightarrow 0.657$), while the meta favorite holds ($R \approx 0.9915$) and corrupt→corrupt N2N stays slightly ahead ($R \approx 0.9922$ on all / top-10%). Favorite wrap-jump on the top-10% subset stays ≈ 1.83 (vs no-bake 2.09). The claim is tiled residual repair, not endpoint-zeroing. **Edge RMSE** is the locked required seam-local secondary (EVAL_PROTOCOL):

Table 7: Multi-family board (20 waveforms; historical prolonged R). ΔR* and ms* vs Dual Cosine only. N2N/seq use train seeds `424242`/`424243`.

Method	R (20-wave)	SNR (dB)	SDR (dB)	wrap-jump	ms*	params	ΔR*
neural favorite	0.977 ± 0.010	35.0 ± 3.2	35.1 ± 3.2	0.90 ± 0.14	2.44	37 489	+0.166
N2N corrupt→corrupt	0.970 ± 0.021	33.9 ± 4.1	34.0 ± 4.1	0.90 ± 0.15	–	53 506	+0.167
seq CNN1D	0.969 ± 0.016	32.2 ± 3.5	32.4 ± 3.5	0.92 ± 0.15	–	3 242	+0.162
N2N sibling-sup.	0.969 ± 0.022	33.7 ± 4.3	33.8 ± 4.3	0.91 ± 0.15	–	53 506	+0.168
poly seam (d=3)	0.966 ± 0.008	31.2 ± 2.1	31.3 ± 2.1	0.91 ± 0.15	1.47	0	+0.148
no-bake	0.965 ± 0.008	30.9 ± 1.9	31.0 ± 1.9	0.91 ± 0.15	0.00	0	+0.147
seq LSTM	0.956 ± 0.037	32.2 ± 5.8	32.3 ± 5.8	0.86 ± 0.15	–	75 746	+0.170
`seam_fir3`	0.955 ± 0.005	28.1 ± 1.2	28.1 ± 1.2	0.46 ± 0.08	0.59	0	+0.136
MLP-on-R	0.932 ± 0.006	24.6 ± 0.9	24.6 ± 0.9	0.64 ± 0.07	2.76	8 604	+0.113
CNN/U-Net-on-R	0.886 ± 0.012	19.5 ± 0.9	19.5 ± 0.9	0.48 ± 0.06	5.01	11 752	+0.069
classic quadratic	0.835 ± 0.012	17.9 ± 1.1	17.7 ± 1.2	0.32 ± 0.05	1.78	0	+0.020
DualCosine	0.814 ± 0.013	16.9 ± 1.2	16.8 ± 1.2	0.32 ± 0.05	1.48	0	0
cosine fade	0.814 ± 0.013	16.9 ± 1.2	16.8 ± 1.2	0.32 ± 0.05	2.64	0	0
crossfade	0.802 ± 0.013	16.7 ± 1.2	16.4 ± 1.2	0.91 ± 0.15	1.70	0	−0.013
hann blend	0.793 ± 0.015	16.1 ± 1.2	15.8 ± 1.2	0.32 ± 0.05	0.57	0	−0.020
soft wide fade	0.729 ± 0.016	14.0 ± 1.1	13.6 ± 1.1	0.63 ± 0.08	3.69	0	−0.082
detrend	0.375 ± 0.040	5.5 ± 1.0	5.4 ± 1.0	0.00 ± 0.00	0.12	0	−0.417
ensemble DC+FIR	0.368 ± 0.040	5.1 ± 1.0	5.0 ± 1.0	0.12 ± 0.02	1.99	0	−0.422

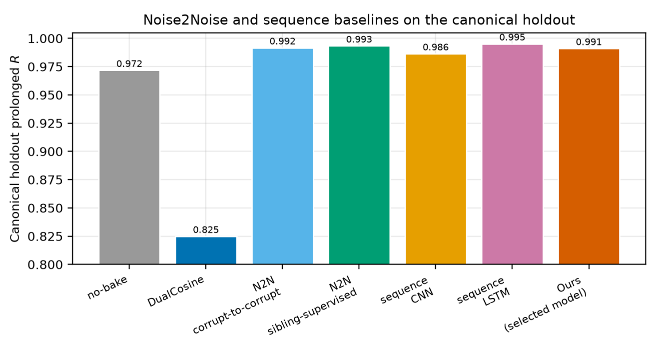

Figure 9: Canonical holdout prolonged R: N2N and seq baselines vs Ours (neural favorite). Train seeds `424242`/`424243` are disjoint from holdout `20260719`.

on the top-10% subset it drops from no-bake 0.095 to favorite 0.021 (N2N corrupt→corrupt also 0.021; DualCosine 0.990). Click energy remains an optional diagnostic.

5.9 Noise2Noise and sequence baselines

How does DenoiseOpt compare to true Noise2Noise-style and lightweight seq restorers on the same generator? Primary corrupt→corrupt N2N (20k Adam steps, 53k params, train seed `424242`) reaches canonical $R \approx 0.9917$, matching the meta favorite within ≈ 0.001. Secondary sibling-supervised N2N reaches $R \approx 0.9933$. A bidirectional LSTM (76k params, 15k steps, train seed `424243`) is the supervised ceiling at $R \approx 0.9952$. A tiny depthwise 1D CNN (3.2k params) reaches $R \approx 0.9864$. Holdout tiles never enter training. Classical DualCosine / `seam_fir3` remain in Table 8. Figure 9 plots these canonical holdout scores beside no-bake, DualCosine, and the neural favorite.

5.10 Factory exports and AKWF cycles

Does the wrap protocol transfer beyond synthetic sine cliffs onto true exports and third-party OA cycles? Table 10 scores ReelSynth-exported factory periods (primary realism, n=25; byte-true factory frames resampled to L=256) and Adventure Kid Waveforms (AKWF) CC0 single-cycle WAVs (secondary, n=24) after close-seam idealization and the same open-wrap cliff. Procedural stand-ins are demoted to tertiary smoke and no longer carry the primary realism claim. On exported factory periods the favorite (R≈0.911) trails no-bake / `seam_fir3` / poly while still beating Dual Cosine (0.883). On AKWF OA cycles the favorite (R≈0.970) matches no-bake and beats Dual Cosine (0.936). Export gaps are listed in Section 9.

5.11 Polynomial baseline and endpoint pinning

Does a cheap classical poly fitter close the residual, and does endpoint pinning show the wrap-jump vs R tradeoff? A seam-local degree-3 polynomial fitter (no learning) reaches canonical $R \approx 0.972$ ($\approx$ no-bake) with wrap-jump essentially unchanged (polynomial results). SSM seq models are deferred; the LSTM remains the supervised seq ceiling. An endpoint-pin control (pinning results) drives wrap-jump to 0 on all / hard subsets while dropping R (all 0.906; hard 0.822). Methods that zero wrap-jump are not the same objective as prolonged R (Limitations).

5.12 Classical virtual-analog seam repairs

Do the cited VA antialias tools [1–3] beat DualCosine on prolonged R when applied as cycle-local seam residuals?

What each operator changes. Figure 10 shows the family on one high-wrap sine+cliff tile (same seed and tile as Fig-

Table 8: Cliff strata on 4,096 holdout tiles (historical prolonged R). Strata by engine wrap-jump.

Method	all R	all jump	top25 R	top25 jump	top10 R	top10 jump
seq LSTM (ceiling)	0.9953	0.865	0.9955	1.637	0.9955	1.810
N2N sibling-sup.	0.9936	0.879	0.9936	1.668	0.9934	1.842
N2N corrupt→corrupt	0.9922	0.870	0.9924	1.647	0.9922	1.821
neural favorite	0.9911	0.869	0.9915	1.645	0.9915	1.825
seq CNN1D	0.9863	0.881	0.9869	1.684	0.9867	1.871
no-bake	0.9715	0.913	0.9713	1.797	0.9667	2.094
`seam_fir3`	0.9611	0.456	0.9461	0.893	0.9429	1.034
DualCosine	0.8193	0.335	0.6864	0.290	0.6574	0.335

n: all 4096, top25 1024, top10 410.

Table 9: Required seam-local secondary on top-10% wrap-jump tiles (n=410): edge RMSE (lower better). Frozen in the cliff-strata data.

Method	top10 edge RMSE
seq LSTM	0.011
N2N sibling-sup.	0.018
neural favorite	0.021
N2N corrupt→corrupt	0.021
seq CNN1D	0.035
no-bake	0.095
`seam_fir3`	0.164
DualCosine	0.990

Table 10: Wavetable-native wrap protocol. Close-seam ideal → open-wrap cliff (seed `20260719`). Mean prolonged R. Primary = ReelSynth export; secondary = AKWF CC0 files.

Method	Export (n=25)	AKWF OA (n=24)
endpoint-pin (mean)	0.939	0.978
`seam_fir3`	0.971	0.975
poly seam (d=3)	0.967	0.970
no-bake	0.967	0.969
neural favorite	0.911	0.970
DualCosine	0.883	0.936
N2N corrupt→corrupt	0.869	0.974
seq CNN1D	0.839	0.962
seq LSTM	0.769	0.948

ure 1). **BLIT/BLEP** (Stilson/Smith lineage) treats the wrap as a value step. It subtracts a band-limited step residual (here a raised-cosine BLEP-family lobe scaled by the measured wrap jump) so the tiled click spectrum is softened near the seam. **PolyBLEP** (Nam et al.) is a cheap polynomial surrogate of that step residual with fractional-delay support near phase wrap. It edits only a few samples on each side of the discontinuity. **BLAMP** (Esqueda et al.) corrects a slope or corner jump (discontinuity in the first derivative), which matters for triangles, rectification, and hard clipping. On a pure value cliff the slope jump is small, so BLAMP is nearly a no-bake passthrough. **DualCosine** (our classical bake baseline) fades both ends of the stored period with raised-cosine windows. It does not insert a BLEP-style residual. It trades wrap energy for a wide end-fade distortion that prolonged R penalizes on hard cliffs.

Scores. Table 11 reports batch means from the VA results (seed `20260719`, batch 64). Implementation is released in the ReelSynth repository (PolyBLEP matches `osc::va::poly_blep`; BLIT/BLEP uses a raised-cosine step residual; BLAMP applies a polyBLAMP lobe to the wrap slope jump). PolyBLEP reaches mean $R \approx 0.929$ ($\Delta R \approx +0.104$ vs DualCosine) but trails no-bake / `seam_fir3` / the meta favorite. BLIT/BLEP nearly zeros wrap-jump (0.004) while dropping mean R to 0.868 and collapsing on hard cliffs ($R \approx 0.712$). BLAMP is near-no-bake on this value-cliff geometry (mean $R \approx 0.972$). On the annotated severe tile in Figure 10, tile R is lower still (PolyBLEP $\approx$ 0.869, BLIT/BLEP $\approx$ 0.703, DualCosine $\approx$ 0.603, BLAMP $\approx$ engine), which matches the hard-cliff strata. Honest reading: these classical tools are scored here, not only cited. They beat DualCosine on mean R in places. None matches the favorite on prolonged R.

5.13 When search helps vs classical methods

Per-cycle ΔR on export, AKWF, Rust `sound_bench`, and sine+cliff holdout (transfer results): favorite win-rate vs Dual Cosine is high on sine cliffs (1.00) and AKWF (0.88), and moderate on export (0.64) / Rust (0.75); win-rate vs no-bake is high only on sine cliffs (0.92) and collapses on export (0.24) / Rust (0.20). Search value is highest on hard cliffs and when compact bake graphs are needed without engine→ideal labels at search time. Export / Rust gaps vs no-bake are listed in Section 9.

5.14 Transfer to Rust engine tiles

Table 12 scores the Python-trained favorite on 20 Rust-exported tiles. The favorite beats Dual Cosine (mean ΔR+0.063, Wilcoxon $p\approx$0.009) but trails no-bake and `seam_fir3` on absolute R. The primary SOTA claim remains the Python multi-family matrix.

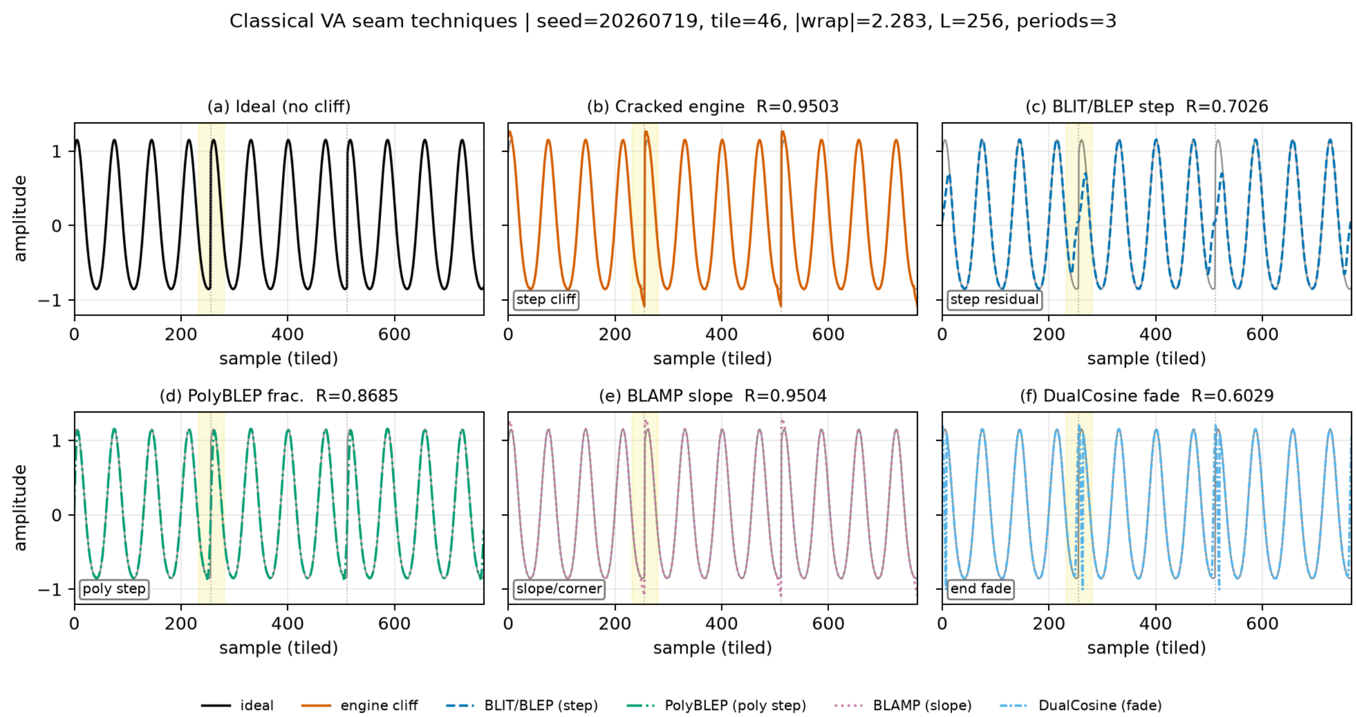

Figure 10: Classical VA seam techniques on one cracked sine+cliff tile (seed `20260719`, tile 46, $|wrap|$=2.283, L=256, three tiled periods). Yellow band marks the first seam neighborhood. Panel tile R (1 = best): (a) ideal (unscored sibling). (b) cracked engine R=0.9503. (c) BLIT/BLEP step residual R=0.7026. (d) PolyBLEP fractional-delay residual R=0.8685. (e) BLAMP slope/corner residual R=0.9504 (near no-bake on value cliffs). (f) DualCosine end fade R=0.6029. Colorblind-safe hues with distinct linestyles for B&W print. Batch means appear in Table 11. Accessibility: six-panel comparison of ideal, cracked, BLIT/BLEP, PolyBLEP, BLAMP, and DualCosine.

Table 11: Classical VA seam repairs on the residual protocol (seed `20260719`). Canonical holdout: R, SNR/SDR, wrap-jump, edge RMSE. Cliff top-10%: R (n=410). Multi: mean R over 20 generative waveforms.

Method	R	SNR	SDR	jump	edge	top10 R	multi R
BLAMP	0.9718	32.6	32.7	0.873	0.080	0.9667	0.965
no-bake	0.9718	32.6	32.7	0.873	0.080	0.9667	0.965
`seam_fir3`	0.9610	29.6	29.6	0.438	0.112	0.9429	0.955
PolyBLEP	0.9288	25.9	25.8	0.102	0.205	0.8641	0.924
BLIT/BLEP	0.8680	20.4	20.3	0.004	0.365	0.7116	0.860
DualCosine	0.8249	18.0	17.8	0.292	0.506	0.6574	0.814

Latency (canonical, CUDA): PolyBLEP $\approx$ 1.43 ms/batch; DualCosine $\approx$ 1.06; BLIT/BLEP $\approx$ 123; BLAMP $\approx$ 54. Frozen in the VA results.

Table 12: Rust `sound_bench` transfer (n=20). Prolonged R; ΔR vs Dual Cosine.

Method	R (20 Rust tiles)	ΔR vs Dual Cosine
no-bake	0.928 ± 0.084	+0.166
`seam_fir3`	0.896 ± 0.077	+0.134
neural favorite	0.825 ± 0.121	+0.063
classic quadratic	0.788 ± 0.110	+0.026
Dual Cosine / cosine fade	0.762 ± 0.118	0

5.15 Vibrato spectrograms

Do wrap-seam differences stay visible under slow vibrato, beyond static tiled periods? Figure 15 renders the same holdout cracked period, DualCosine bake, and refit **Ours (hybrid GA–PPO)** champion under shared sinusoidal vibrato ($\pm 3\%$ depth at 5 Hz around 220 Hz). Panels (a–c) show magnitude spectrograms; (d–e) show DualCosine−no-bake and Ours−no-bake differences; (g) reports cycle-local prolonged R alongside a modulated-playback residual under the same pitch envelope. On the annotated high-wrap tile, Ours retains high cycle R and high modulated R, while DualCosine remains the weakest classical comparator. This is objective spectro-

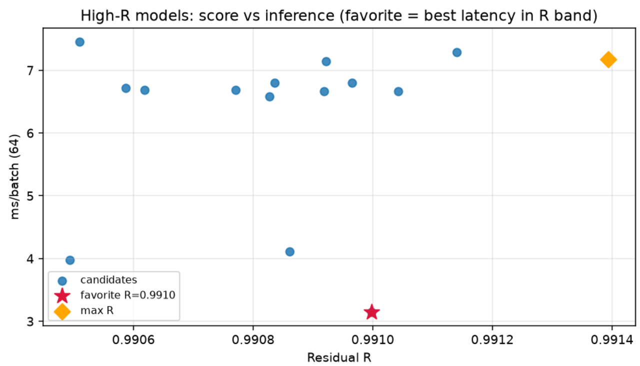

Figure 11: High-R fitted cells: residual vs CUDA latency. Star: favorite.

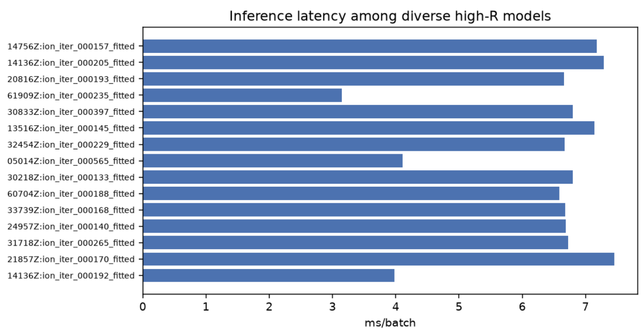

Figure 12: Inference latency for high-R fitted models (CUDA, batch 64).

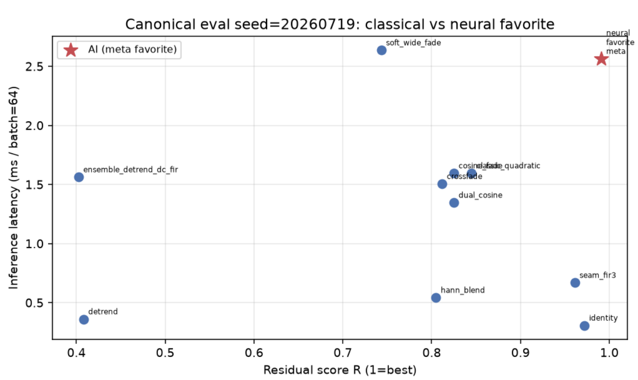

Figure 13: Canonical holdout: classical methods and favorite in the R vs ms/batch plane.

gram / residual evidence only. It is not a formal listening study. Protocol: `bench_vibrato_spectrogram.py`; artifacts: vibrato folder.

5.16 Downloadable audio samples

As an audible companion to Figure 6, we release five fixed-pitch (A4, 3 s, 44.1 kHz mono) wavetable clips for no-bake, DualCosine, and Ours-healed periods from the matched-suite holdout (hear-sample WAVs; rebuild: `export_meta_hear_samples.py`). Figure 16 shows

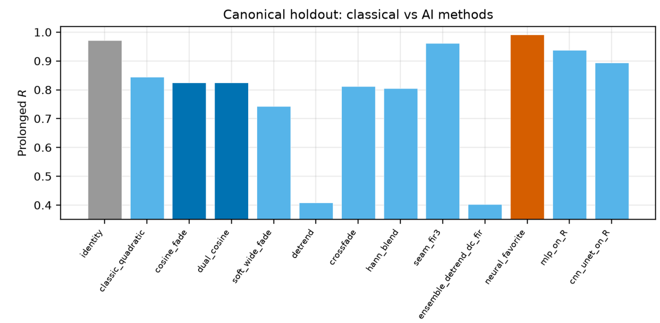

Figure 14: Canonical holdout ranking on prolonged R (left) and latency (right).

short waveform excerpts from those WAVs with per-clip absolute R. Readers can download and listen; we do *not* report formal MOS/MUSHRA or claimed A/B listening scores.

5.17 Factory and Adventure Kid waveform examples

Beyond the synthetic sine+cliff narrative, Figure 17 shows a compact gallery of true ReelSynth-exported factory periods (one mid-morph frame per bank) and Adventure Kid Waveforms (AKWF) CC0 single-cycle files already used by the wrap protocol. Scored means remain in Table 10 / wavetable results; this panel is diversity evidence only. No new NAS was run for the gallery.

5.18 Transfer to bearings, ECG, and synthetic cycles

Setup reminder. Table 13 reports whole-curve prolonged residual R after training/searching DenoiseOpt (`hybrid_lstm`, 250 outer iterations, seed `1902771841`) on each transfer board from Section 4.2 (n=256 periods, L=256). Columns are: CWRU/MFPT bearing revolutions, MIT-BIH/PTB-XL ECG beats, and synthetic CNC/PMU pilots. This pilot uses the historical prolonged-R FitCell objective (frozen results_table.json); it is *not* a v10.1 R_{blend} retag. Domain-trained Noise2Noise quality scores were not run on these boards (latency Table 19 reports zero-shot wavetable N2N timing only). Deep SOTA rows remain *not executed* (Table 14). Artifacts: signal_heal_transfer.

Takeaways. Under prolonged R, Ours leads every classical-board column in Table 13, including Dual Cosine on every domain. On CWRU/MFPT, Dual Cosine and fades trail no-bake; Ours clears both by a large margin. On MIT-BIH and PTB-XL, endpoint-pin is a strong classical ECG row, but Ours still leads. Synthetic CNC shows the largest classical collapse with Ours near 0.992. Domain-trained Noise2Noise quality was not run on these boards (zero-shot timing in Appendix B). Primary claim remains cycle-local wavetable seam restoration.

5.19 Listening protocol (no formal listening study)

Downloadable WAVs (hear-sample WAVs) and vibrato spectrograms support informal A/B inspection (Sections 5.15–5.16). Formal MOS / MUSHRA scores were not collected. Until a formal study runs, perceptual claims remain qualitative and secondary to residual scores. See Section 9.

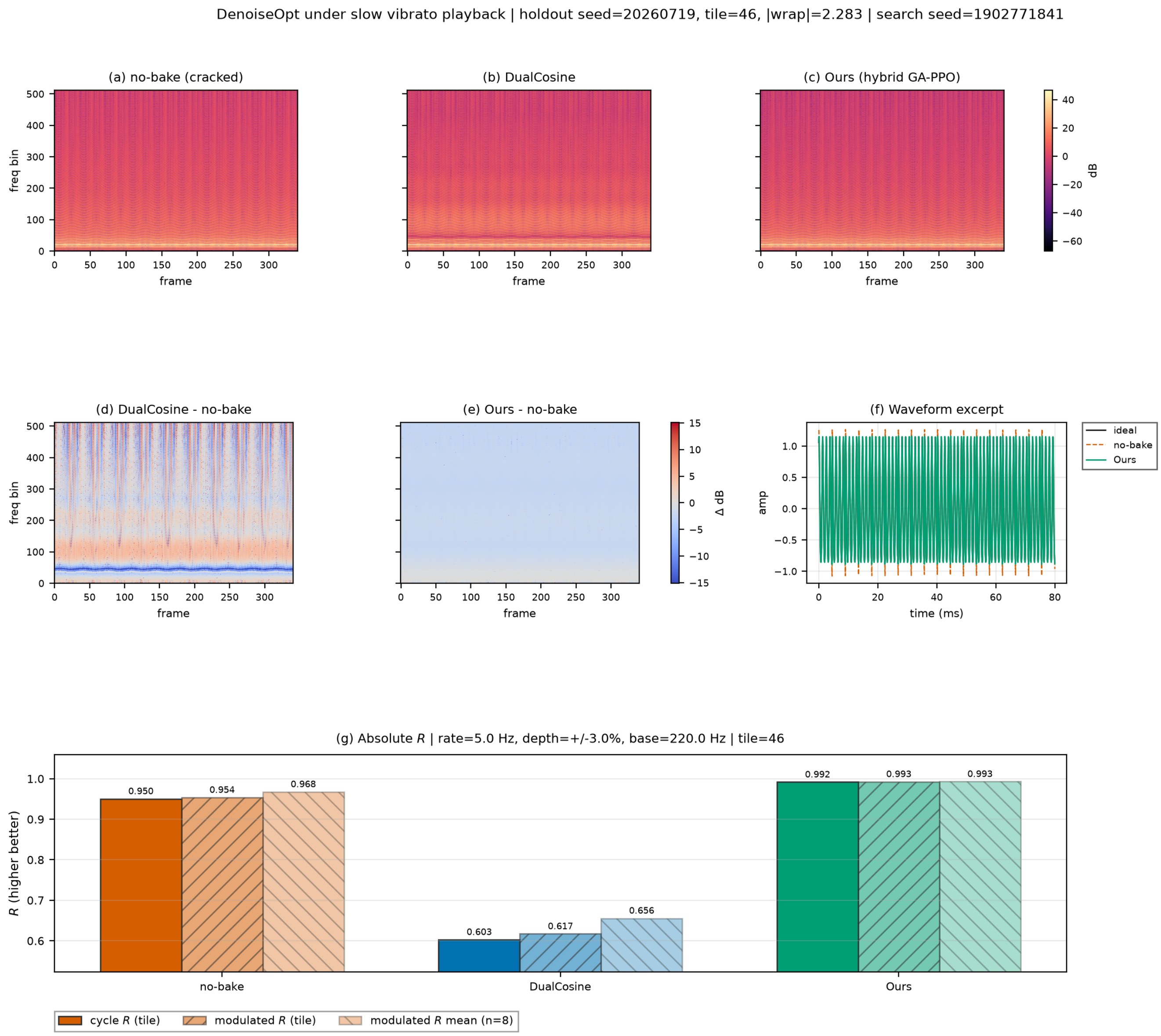

Figure 15: Slow-vibrato spectrogram eval on a high-wrap holdout tile (seed `20260719`). (a–c) Magnitude spectrograms for no-bake, DualCosine, and Ours. (d–e) Spectrogram differences vs no-bake. (f) Short waveform excerpt. (g) Absolute R: cycle-local prolonged residual, modulated-playback residual on the same tile, and mean modulated R over top-wrap tiles. Accessibility: Okabe–Ito hues; difference maps use diverging coolwarm. Formal human A/B scores are out of scope.

6 Discussion

Main story under R_{blend}. The locked protocol does not crown the hybrid champion over Noise2Noise on the holdout (0.9697 vs 0.9750). Dual Cosine on that board sits near 0.541. The useful reading is therefore not "searched bake beats every neural baseline." It is: compact searchable bake cells that crush classical fades, stay competitive with Noise2Noise on multi-family means (13/20 wins), and hold on hard cliffs where Dual Cosine collapses (Table 4, Section 5.8). Older whole-curve numbers near $R \approx 0.991$ belong in the historical boards only (Limitations).

Which families stay hard. Across large procedural benches, wrap error is uneven. Families such as `nonlinear`, `combo`, `triple_mix`, and `extreme_overlay` dominate the residual budget; `harmonic_fft` and `am_fm` are easier. Mean scores hide that structure.

Search near the ceiling. Under near-ceiling residuals, further gains are small in absolute units and sensitive to reward shaping and branch weights. Plateau adaptation (deeper cells, denser MoE graphs, higher GA/PBT weight) is the operational response inside one campaign. Matched outer-loop bake-offs and HP probes sit in Appendix B.

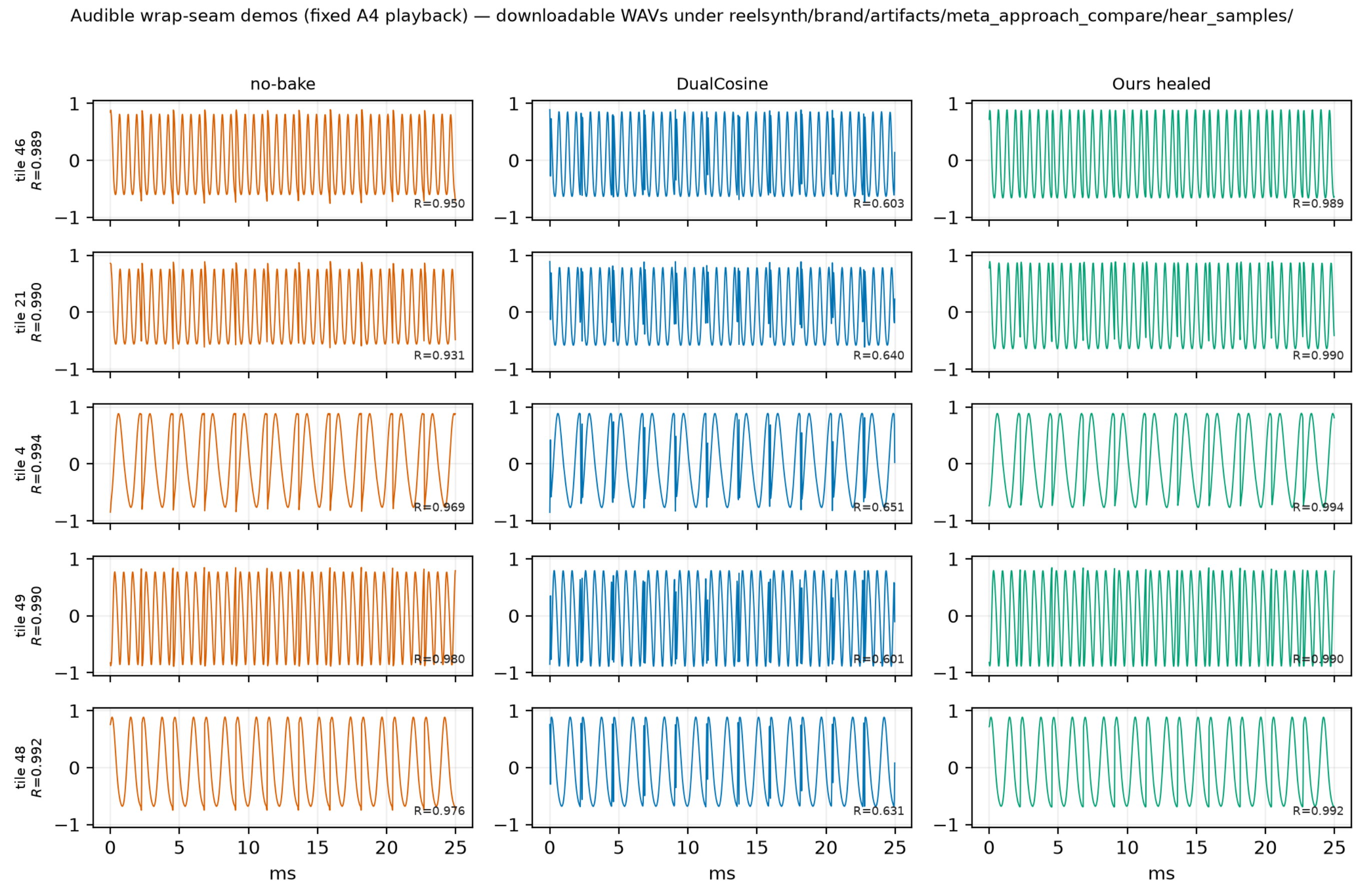

Figure 16: Hear-sample panel from the online WAV pack (hear-sample WAVs; five holdout tiles; no-bake / DualCosine / Ours). Waveforms show the first ≈ 25 ms of each clip; R annotations are cycle-local prolonged residuals vs the ideal sibling. Not a formal listening study.

Table 13: Wrap-heal transfer under prolonged residual R (higher better; historical whole-curve score). DenoiseOpt trains/searches `hybrid_lstm` for 250 iterations per domain (seed `1902771841`). Boards: CWRU/MFPT bearing revolutions, MIT-BIH/PTB-XL ECG beats, synthetic CNC/PMU pilots (n=256 periods each). Classical board only; domain-trained N2N quality not executed; no invented deep-SOTA scores.

Method	CWRU	MFPT	MIT-BIH	PTB-XL	synth CNC	synth PMU
Ours (`hybrid_lstm`)	0.9623	0.9411	0.9565	0.9336	0.9918	0.9915
no-bake	0.8451	0.8657	0.7495	0.7503	0.4499	0.9356
endpoint-pin	0.5054	0.5446	0.8650	0.8569	0.4096	0.9442
`seam_fir3`	0.4483	0.5726	0.7662	0.7462	0.5344	0.9119
DualCosine	0.4608	0.4830	0.4137	0.4769	0.4943	0.8272

Evidence beyond static tiles. Vibrato spectrograms and downloadable hear WAVs support informal inspection (Sections 5.15–5.16). Sci/eng wrap transfer is a classical-board stress test (Section 5.18). Perceptual and scope limits are collected in Section 9.

7 Tooling and AI use

Open literature was gathered with local OpenAlex / Crossref / arXiv providers; every bibliography entry has a downloadable OA PDF (literature PDFs). Section planning used research-assistant tools as scaffolding only. Claims, numbers, and citations were checked against measured JSON artifacts and code. Fitting, residual scoring, baselines, and hybrid GPU search are conventional Rust / PyTorch with deterministic logs. AI did not invent experimental numbers. Audible demos: hear-sample WAVs.

8 Broader impact and reproducibility

Broader impact. This work targets cycle-local wavetable seam repair for virtual-analog / software-instrument work-

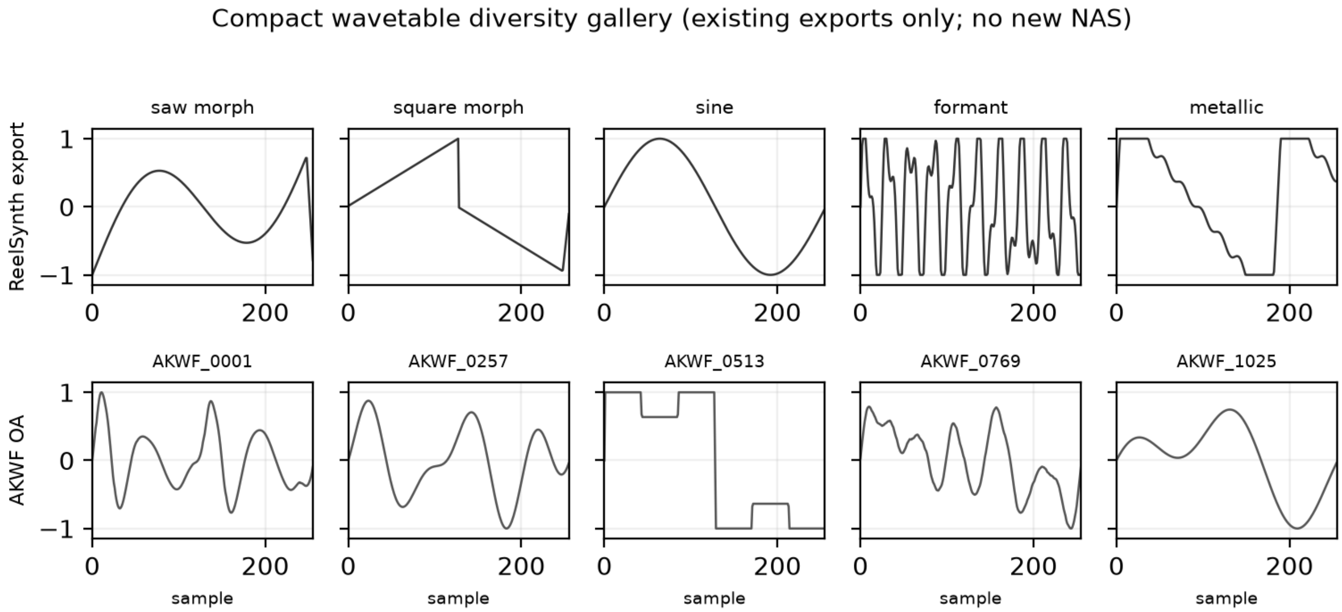

Figure 17: Compact wavetable diversity gallery from existing exports (one mid-morph dry frame per unique ReelSynth factory bank, top; matched AKWF CC0 cycles, bottom). No new NAS for this panel. Quantitative wrap scores: Table 10.

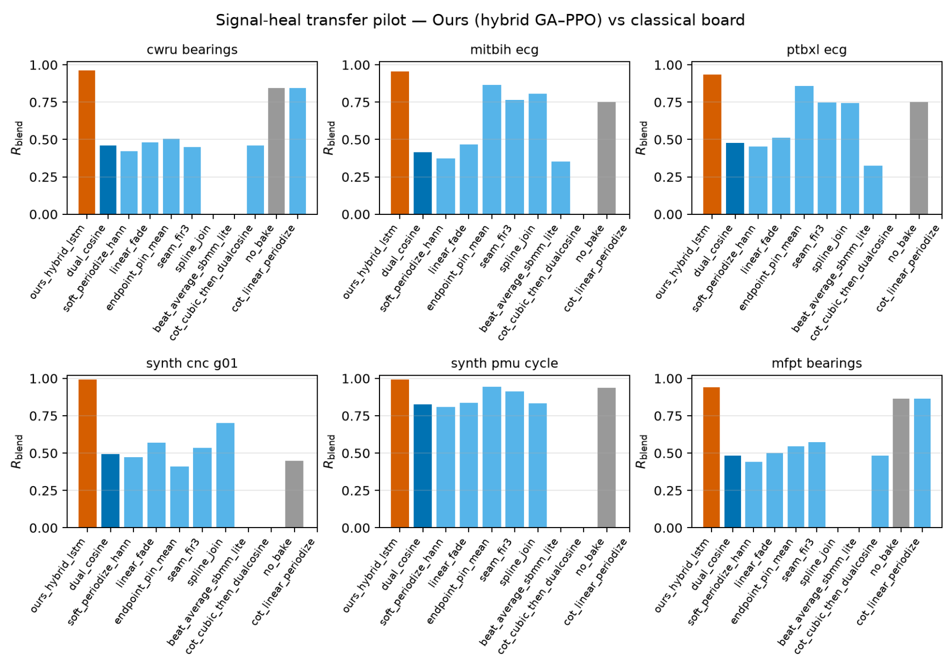

Figure 18: Transfer classical-board summary (holdout-refit prolonged R; grouped bars for Table 13). Not a deep-SOTA claim; domain-trained N2N quality not run; see Table 14.

flows. Misuse potential is low. The method does not enhance speech for surveillance, does not remove watermarks from protected media, and operates only on short synthetic periods. Training and search use procedural sine+cliff generators. No private studio stems are required for the reported tables. The main resource cost is GPU time on a consumer RTX 3090 (order of hours for the 5k-gate freeze, Table 17). Authors should not over-claim speech enhancement or production mastering quality from these metrics.

Reproducibility. Holdout seed `20260719`. Hybrid search seed `1902771841`. Primary code: `https://github.com/reeldemo/reelsynth` (search runner: `overnight_gpu_rl_arch.py`;

Table 14: Deep / real-data transfer status under this residual protocol. No invented scores.

Item	Status
Domain-trained N2N quality	*not executed* (zero-shot WT N2N latency only)
Cycle-GAN (ECG)	*not executed* (no weights under this residual)
BeatDiff	*not executed*
Paderborn KAt deep	*not executed* (registration / multi-GB)
Full PTB-XL corpus	deferred (`records500 *_hr` subset, n=256)
Real KIT CNC / IEEE PMU	deferred (login walls; synth pilots ran)
Formal MOS/MUSHRA	deferred (Section 5.19)

SOTA bench: `bench_sota_matrix.py`; VA seam: `va_seam_blep.py`). Paper companion: https://github.com/reeldemo/denoise-opt-meta. Dependencies for the CUDA benches: Python 3.12, PyTorch 2.6+CUDA 12.4, NumPy/matplotlib. Rust/`cargo` builds the separate engine hardness probe. Frozen numbers live under brand/artifacts (e.g. `sota_matrix.json`, `va_seam_blep.json`). PESQ/STOI are omitted for non-speech cycles (domain mismatch). MUSHRA was not run.

8.1 Funding

This research received no external funding.

8.2 Author contributions (CRediT)

Julian M. Kleber: Conceptualization, Methodology, Software, Investigation, Formal analysis, Data curation, Writing (original draft), Writing (review and editing), Visualization.

8.3 Data availability

Frozen evaluation artifacts (JSON matrices, search history, fitted champions) are published with the companion repositories: https://github.com/reeldemo/reelsynth (brand/artifacts) and https://github.com/reeldemo/denoise-opt-meta (artifacts and this paper folder's figures). All reported tables regenerate from those files plus the scripts named above. No private studio recordings were used.

8.4 Competing interests

None declared.

9 Limitations

We collect honesty notes here rather than repeating them through Results and Discussion.

- **Not a perceptual score.** R_{blend} and J use a procedural no-cliff sibling for seams and an engine-identity prior for the mid-cycle body. They do not measure listening quality. No formal MOS or MUSHRA was run. Downloadable WAVs and vibrato spectrograms support informal A/B only (Section 5.19). PESQ and STOI are excluded: primary cycles are non-speech wavetable tiles.
- **Locked story vs older whole-curve R.** Primary claims use R_{blend} under the v10.1 lock (Table 4). Many secondary tables and figures still report historical whole-curve prolonged R (canonical board, cliff strata, matched outer loops, VA board). Those boards are labeled as such; they must not overwrite the holdout gate reading ($R_{\text{blend}}\approx$0.9697 for Ours vs 0.9750 for Noise2Noise).
- **Where Noise2Noise was not run.** Domain-trained Noise2Noise quality was not executed on sci/eng transfer boards (only optional zero-shot timing in Appendix B). Rust `sound_bench` scores classical rows and the favorite only.
- **Domain shift on exports.** On Factory+FX / Rust tiles the meta favorite can trail no-bake or FIR while still beating Dual Cosine. We do not claim universal factory superiority.
- **Search geometry.** Search scores Python generator draws. The Python multi-family matrix is primary; Rust export is secondary. Per-trial width is capped; Demucs-/diffusion-scale stacks were screened out.
- **No wrap-closure theorems.** Endpoint-pin can zero wrap-jump while residual falls. We claim tiled residual repair toward $r^\star$, not endpoint closure or search convergence.
- **Transfer scope.** Sci/eng wrap transfer is a classical-board pilot, not deep CWRU/ECG SOTA (Table 14).
- **Hyperparameter probe.** The $\pm$50% one-at-a-time probe (Appendix B) is not a full re-search of Table 2.

9.1 Future work

A follow-up can treat which sounds fail as the primary object: stratified residual heatmaps over procedural families, cliff-size conditioning, and family-specialist cells under the same outer loop. That paper would cite the present hybrid protocol as baseline and ask whether further gains are mainly architecture or coverage over hard families. Formal listening (even a small author-plus-colleague preference note on high-wrap tiles) remains the clearest missing perceptual check.

10 Conclusion

DenoiseOpt casts wavetable wrap discontinuity as cycle-local seam repair under blended residual R_{blend} and latency-aware objective J. A hybrid GA+PPO(+PBT) outer loop searches compact bake cells without paired clean targets at search time. Under the locked protocol, Noise2Noise is the neural gate and Dual Cosine is the classical fade baseline. The hybrid freeze reaches $R_{\text{blend}}\approx$0.970 on holdout but does not clear Noise2Noise ($\approx$0.975); Dual Cosine sits near 0.541. On multi-family boards Ours edges Noise2Noise on mean score; on hard cliffs classical fades collapse while searched cells hold. The contribution is robust searchable seam cells for DSP / synthesis, not speech-enhancement or diagnosis SOTA. Code: https://github.com/reeldemo/reelsynth, https://github.com/reeldemo/denoise-opt-meta.

Algorithm 1 ResidualScore

Require: Ideal cycle $r^\star$, baked cycle y, periods N

$I \leftarrow \mathrm{Tile}(r^\star, N)$
$Y \leftarrow \mathrm{Tile}(y, N)$
$r_{\mathrm{rms}} \leftarrow \mathrm{RMS}(Y - I)$
$i_{\mathrm{rms}} \leftarrow \max(\mathrm{RMS}(I), \varepsilon)$
return $\mathrm{clamp}(1 - r_{\mathrm{rms}}/i_{\mathrm{rms}}, 0, 1)$

Algorithm 2 DualCosine baseline

Require: Engine cycle x

Fade both ends of x with raised-cosine windows of width W
return ResidualScore$(r^\star, x_{\mathrm{faded}}, N)$

Algorithm 3 No-bake (passthrough) control

Require: Engine cycle x (open-wrap cliff present)

return ResidualScore$(r^\star, x, N)$ ▷ no operator: $x \mapsto x$

Acknowledgments

Compute for the CUDA hybrid search campaign used a local NVIDIA GeForce RTX 3090 workstation. Open literature was retrieved via the Klaut Research Gateway under open-access PDF constraints. No external collaborators contributed text, code, or experimental numbers for this preprint.

A Algorithms

Companion pseudocode for the residual score, classical controls, FitCell, and hybrid outer-loop branches. Canonical narrative lives in Section 3; these listings mirror the pseudocode notes and the CUDA search runner.

B Supplementary tables and figures

Material moved here for the conference-length reading path: matched outer-loop bake-off, branch/compute freezes, hyperparameter probe, transfer latency, and search-monitor plots. Primary claims remain in Sections 5–5.18.

B.1 Matched-budget outer-loop comparison

Under a matched 5,000-evaluation budget with LSTM and xLSTM in the searchable bake vocabulary, how do Random NAS, Cont. CMA-ES, Arch REINFORCE, Aging evolution, and TPE Bayes NAS compare to Ours (hybrid GA+PPO)? Table 15 reports champion prolonged R, ΔR vs Dual Cosine, wall hours, and whether the champion graph used an LSTM or xLSTM block. This is an outer-loop bake-off under historical whole-curve prolonged R. Noise2Noise is not an outer-loop method here (see Tables 4 and 7). At the completed gate, Ours leads on absolute champion R ($R \approx 0.99146$, $\Delta R \approx +0.17488$ vs Dual Cosine ≈ 0.81658; ≈ 8.46 h), with Aging evolution second ($R \approx 0.99079$). Neither Ours nor Aging evolution selected an LSTM block in the champion graph. Source: matched-budget results (search seed `1902771841`).

Algorithm 4 MoE soft gating over bake experts

Require: Cycle x, expert cells $\{E_k\}_{k=1}^{K}$, gate logits $g \in \mathbb{R}^K$

$w \leftarrow \mathrm{softmax}(g)$
$y \leftarrow \sum_{k=1}^{K} w_k E_k(x)$
return y

Algorithm 5 FitCell (inner loop)

Require: Architecture cfg, fit steps T, batch size B

cell $\leftarrow$ SeamCell(cfg); opt $\leftarrow$ Adam(cell.params)
prev $\leftarrow$ null; patience $\leftarrow 0$
for $t = 1, \ldots, T$ **do**
  ideal, eng $\leftarrow$ MakeBatch(B)
  out $\leftarrow$ ApplyOps(eng, cell, cfg.ops)
  $R \leftarrow$ mean ResidualScore(ideal, out)
  loss $\leftarrow 1 - R$
  if finite(loss) **then** AdamStep(opt, loss)
  end if
  if prev $\neq$ null and $|\mathrm{prev} - R| / \max(|\mathrm{prev}|, \varepsilon) < 10^{-4}$ **then**
    patience $\leftarrow$ patience $+1$; **if** patience ≥ 3 **then break**
  else
    patience $\leftarrow 0$
  end if
  prev $\leftarrow R$
end for
return cell, R

Algorithm 6 GA tournament step

Require: Population P, tournament size k, crossover rate p_c, mutation rate p_m

Sample k genomes; parents $\leftarrow$ top-2 by R
child $\leftarrow$ Crossover(parents) with prob. p_c, else clone elite
Mutate(child) with rate p_m (ops, depth, cell type, θ)
Fit and evaluate child; insert into P (replace worst if full)
return updated P

B.2 Ablations and compute

Table 16 reports branch-best freezes at the 5k gate from one hybrid history, plus isolated 150-iteration re-runs on the same search seed. Table 17 summarizes campaign compute.

B.3 Hyperparameter sensitivity

One-at-a-time $\pm 50\%$ probe of Table 2 knobs at 500 outer iterations (seed `1902771841`). Not a full 5k re-search. Default champion $R \approx 0.9888$; largest $|\Delta R|$ among completed arms ≈ 0.0123 (50% lower learning rate). All 11/11 arms completed.

B.4 Transfer inference latency

Table 19 and Figure 22 pair with the prolonged-R transfer board (Table 13). N2N uses the wavetable checkpoint zero-shot; domain-trained N2N quality was not run.

Algorithm 7 PPO architecture update

Require: Actor–critic π_ϕ, clip ε, experience buffer of mutate actions
Sample action a editing ops / hyperparameters from π_ϕ
Fit proposed cfg; observe centered advantage $\rho = R - R_{\text{DualCosine}}$ ▷ objective remains maximize R
Advantage $\hat{A}$ from critic; ratio $r_\phi = \pi_\phi(a)/\pi_{\phi_{\text{old}}}(a)$
Update ϕ with clipped surrogate $\min(r_\phi \hat{A}, \text{clip}(r_\phi, 1-\varepsilon, 1+\varepsilon)\hat{A})$
return updated π_ϕ

Algorithm 8 PBT exploit–mutate

Require: Population P, exploit quantile q, mutate scale σ
Rank P by R; for each bottom member, copy hyperparameters from a top-q elite
Perturb copied hyperparameters by multiplicative noise of scale σ
Refit briefly; write back into P
return updated P

Algorithm 9 HybridMetaSearch

Require: Iteration budget T_{out}, population size n
$P \leftarrow \text{InitPopulation}(n)$; $\pi \leftarrow \text{ActorCritic}()$; champ $\leftarrow$ null; boredom $\leftarrow 0$
for $it = 1, \ldots, T_{\text{out}}$ **do**
    branch $\leftarrow \text{Rotate}(\text{ppo}, \text{ga}, \text{pbt}, \text{nas}, \text{combo})$
    cfg, hp $\leftarrow \text{Propose}(\text{branch}, P, \pi)$
    cell, $R_{\text{fit}} \leftarrow$ FitCell(cfg, hp.fit_steps)
    $R_{\text{eval}} \leftarrow$ EvalCell(cell)
    $R \leftarrow \frac{1}{2}R_{\text{fit}} + \frac{1}{2}R_{\text{eval}}$ (+ optional depth/MoE bonus)
    UpdatePopulation(P, cfg, R); PPOUpdate(π, centered $\rho = R - R_{\text{DualCosine}}$)
    if $R >$ champ.R **then** champ $\leftarrow$ (cfg, cell, R); boredom $\leftarrow 0$
    else boredom $\leftarrow$ boredom $+1$
    end if
    if boredom ≥ 1000 **then** PlateauAdapt; boredom $\leftarrow 0$
    end if
end for
return champ

B.5 Search-monitor figures

C Transfer artifact locations

Full cross-domain wrap-heal transfer protocol, classical-board tables, and deep-model status rows are in Sections 4.2–5.18 of the main text. Algorithms for the residual score and hybrid loop remain in Appendix A. Raw JSON and figures: signal_heal_transfer.

Table 15: Search methods at a matched 5,000-evaluation budget (search seed `1902771841`; historical prolonged R). Champion R is measured against the ideal sibling; ΔR is reported relative to DualCosine (classical fade), not Noise2Noise. Wall time is per-method CUDA time. LSTM and xLSTM indicate whether the champion graph included that block.

Method	R @ 5k	ΔR vs DC	h	LSTM	xLSTM
Random NAS	0.97816	+0.16157	5.72	no	no
Cont. CMA-ES	0.98468	+0.16810	2.28	no	no
Arch REINFORCE	0.98247	+0.16588	25.93	no	yes
Aging evolution	0.99079	+0.17421	15.02	no	yes
TPE Bayes NAS	0.98106	+0.16448	11.18	yes	no
Ours (hybrid GA–PPO)	0.99146	+0.17488	8.46	no	no

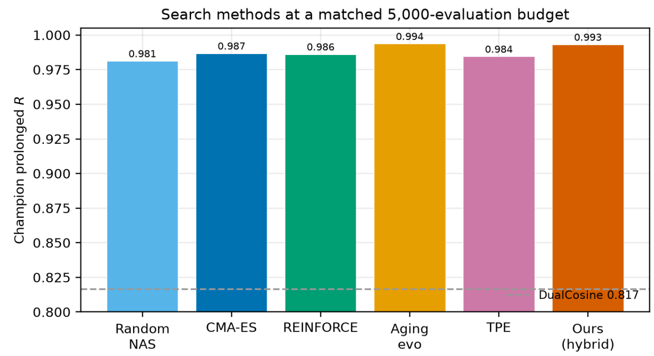

Figure 19: Matched 5,000-evaluation search methods (seed `1902771841`). Champion prolonged R; dashed line is Dual Cosine.

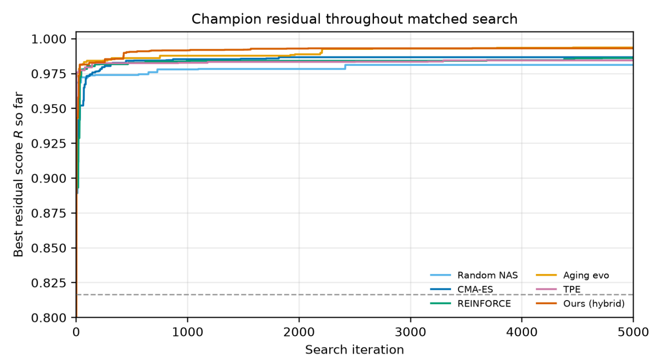

Figure 20: Champion residual R over a matched 5,000-evaluation budget. Dashed line: Dual Cosine.

Table 16: Ablations. Left: best score by branch in the recorded search run (final iteration ≈6,922). Right: isolated 150-iteration re-runs (seed `1902771841`). Prolonged R.

Config	Search freeze R	Isolated 150-it R
Full hybrid	0.99093	0.98113
GA-only	0.99016	0.98062
PBT (search branch)	0.99012	n/a
PPO-only	0.98924	0.97932
GA+PPO	n/a	0.98007
Combined branch	0.98854	n/a
NAS branch	0.98677	n/a
Dual Cosine (runner)	0.81658	0.81658

Table 17: Compute budget for the reported 5,000-evaluation search run (RTX 3090, search seed `1902771841`).

Item	Value
GPU	NVIDIA GeForce RTX 3090
Elapsed wall time	≈7.92 h
Clean iterations / arch evals	≈6,922
Population size	12
Final champion R	0.99093
Dual Cosine baseline (runner)	0.81658
Peak VRAM (replay)	≈3.29 GiB

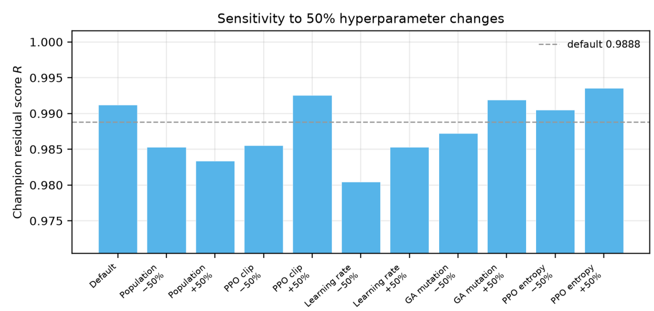

Figure 21: One-at-a-time ±50% HP sensitivity of hybrid champion R at 500 iterations. Blue bar: Table 2 default.

Table 18: One-at-a-time ±50% hyperparameter sensitivity probe (500 outer iterations, seed `1902771841`, sine+cliff FitCell). Champion absolute R is compared with the Table 2 default. This is sensitivity evidence only, not a full 5,000-iteration re-search for each parameter.

Parameter change	Champ. R	ΔR	Iter.
Default	0.98882	+0.00000	500
Population size −50%	0.98136	-0.00746	500
Population size +50%	0.98073	-0.00809	500
PPO clip −50%	0.98159	-0.00723	500
PPO clip +50%	0.99041	+0.00159	500
Learning rate −50%	0.97654	-0.01228	500
Learning rate +50%	0.98138	-0.00744	500
GA mutation rate −50%	0.98334	-0.00548	500
GA mutation rate +50%	0.99002	+0.00120	500
PPO entropy −50%	0.98912	+0.00030	500
PPO entropy +50%	0.99116	+0.00234	500

Table 19: Transfer-domain inference latency (ms/batch). Batch=64; CUDA.

Method	CWRU	MFPT	MIT-BIH	PTB-XL	CNC	PMU
no-bake	0.26	0.26	0.27	0.27	0.28	0.26
Dual Cosine	1.10	1.11	1.12	1.14	1.15	1.14
Ours (hybrid)	3.92	6.61	2.87	1.67	1.69	1.67
N2N (WT to domain)	0.95	0.99	0.98	0.97	0.98	0.97

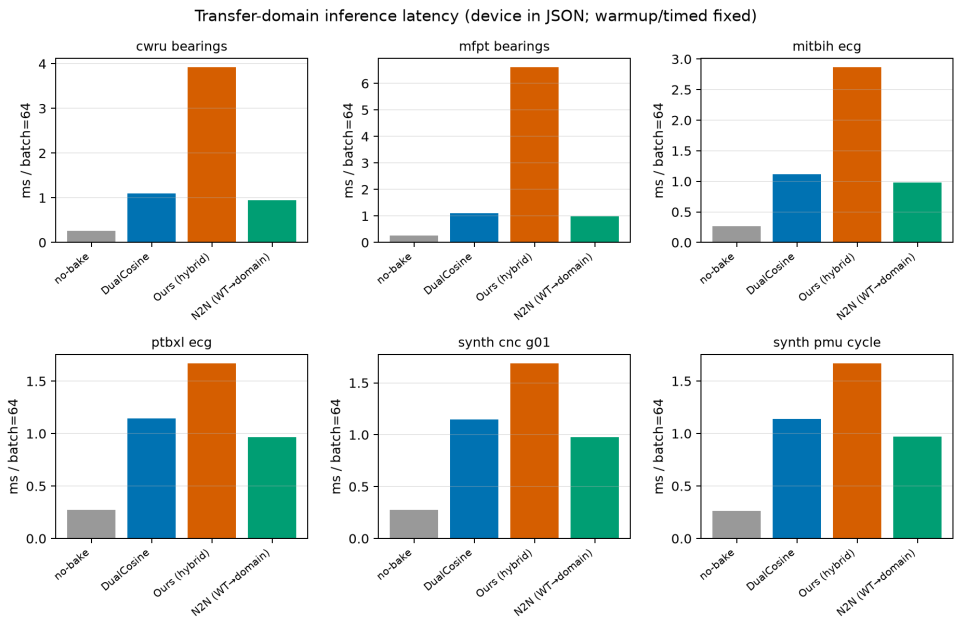

Figure 22: Transfer-domain inference latency (ms/batch). N2N is wavetable-trained zero-shot when available.

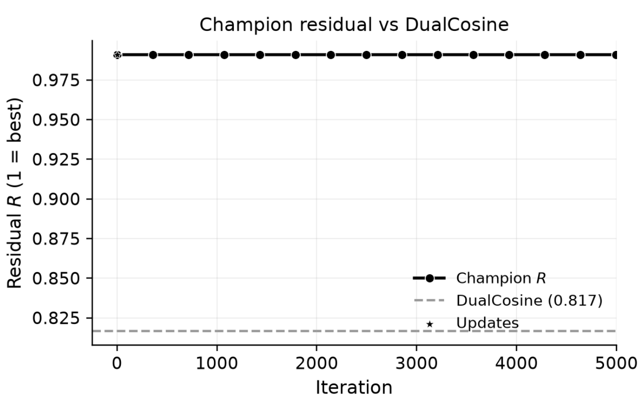

Figure 23: Champion residual R over the first 5,000 search iterations (Dual Cosine dashed).

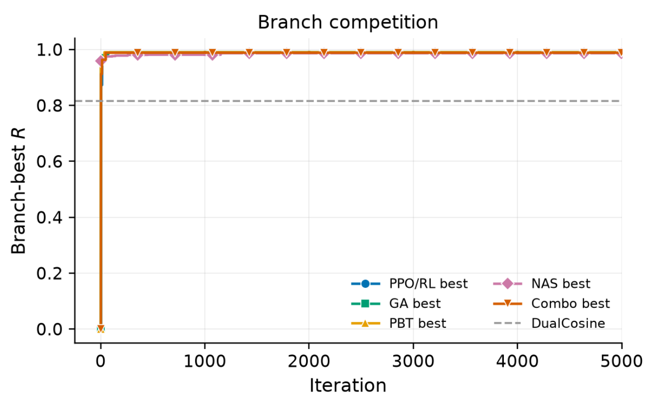

Figure 24: Best residual by search branch over the first 5,000 iterations.

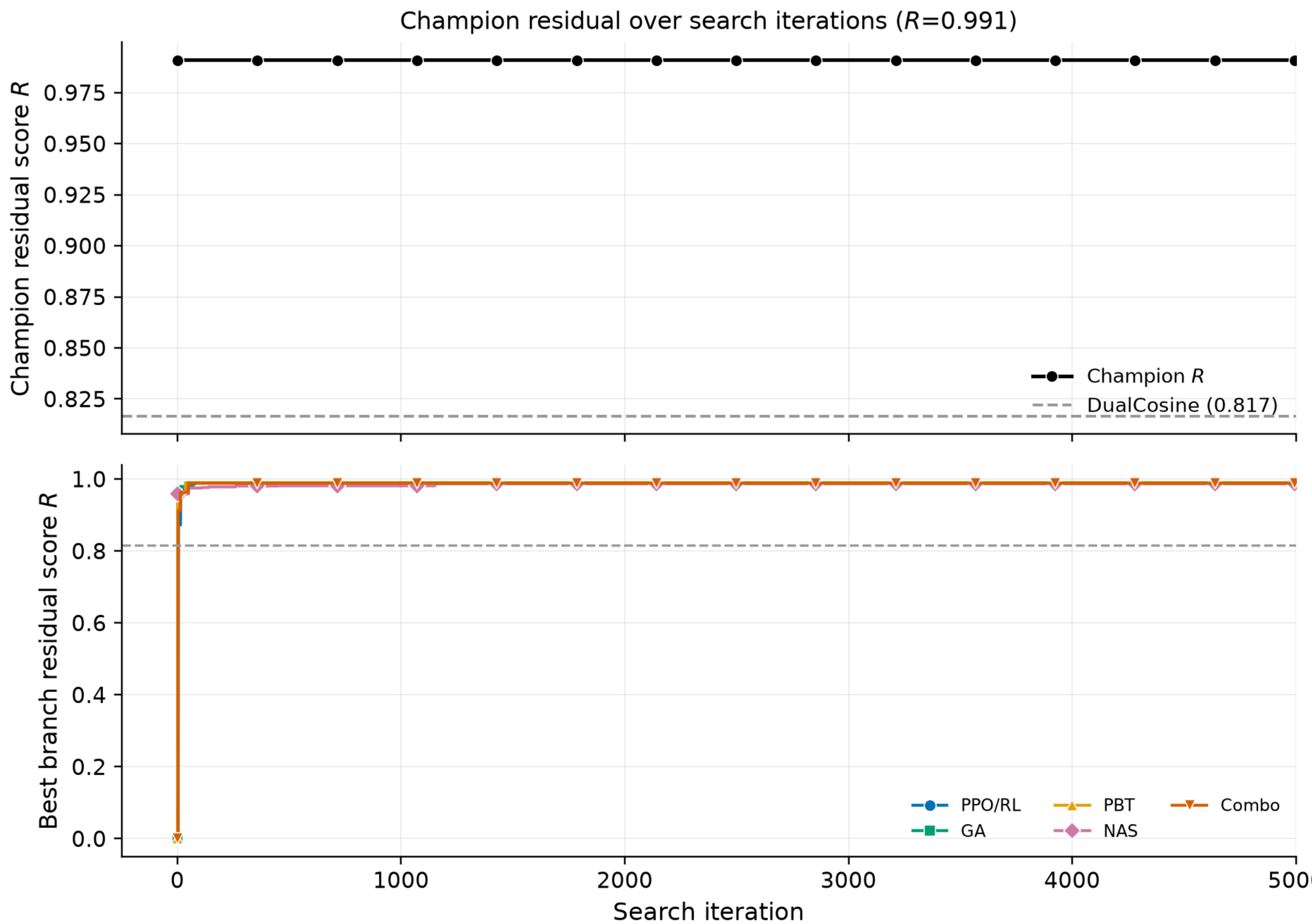

Figure 25: Search progress: champion R (top) and branch bests (bottom).

Figure 26: Per-trial residual by branch with champion trajectory overlaid.

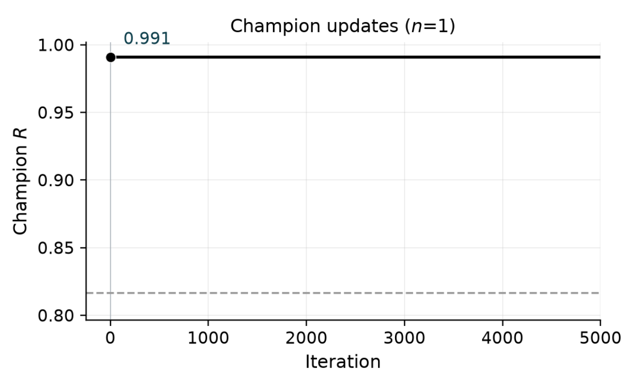

Figure 27: Champion updates over the first 5,000 iterations.

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